0 Data Frame Management

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The goal of this script is to create and clean the necessary dataframes to be used in analysis later. Run this first then move on to other scripts.

Reading CSVs in

Okay, this is the boring part. I need to just load in all my datasets and get everything nice and pretty. I'm going to have to deal with plant names.

Places

Loading in data based on where it was collected

```
Proper_Names <- read_csv(here("EnteredData", "PlantNameChanger.csv"))
```

```
## Rows: 33 Columns: 2
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (2): Old_Species_Name, New_Species_Name
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Ten Mile
TeMi_Coords <- read_csv(here("EnteredData", "TeMi_RelativeCoords.csv"))
```

```
## Rows: 86 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (10): OID_, IN_FID, NEAR_FID, NEAR_DIST, NEAR_RANK, FROM_X, FROM_Y, NEAR...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
TeMi_Dist <- read_csv(here("EnteredData", "TeMi_Distance.csv"))
```

```
## Rows: 1806 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (10): OID_, IN_FID, NEAR_FID, NEAR_DIST, NEAR_RANK, FROM_X, FROM_Y, NEAR...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
TeMi_Locations <- read_csv(here("EnteredData", "TeMi_Location.csv"))
```

```
## Rows: 43 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Comment
## dbl (5): OID_, POINT_X, POINT_Y, POINT_Z, POINT_M
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Zmudowski
Zmud_Coords <- read_csv(here("EnteredData", "Zmud_RelativeCoords.csv"))
```

```
## Rows: 88 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (10): OID_, IN_FID, NEAR_FID, NEAR_DIST, NEAR_RANK, FROM_X, FROM_Y, NEAR...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
Zmud_Dist <- read_csv(here("EnteredData", "Zmud_Distance.csv"))
```

```
## Rows: 1892 Columns: 11
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (11): OID_, OBJECTID, IN_FID, NEAR_FID, NEAR_DIST, NEAR_RANK, FROM_X, FR...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
Zmud_Locations <- read_csv(here("EnteredData", "ZmudLocations.csv"))
```

```
## Rows: 44 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Comment
## dbl (5): OID_, POINT_X, POINT_Y, POINT_Z, POINT_M
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Guadalupe Nipomo
GuNi_Coords <- read_csv(here("EnteredData", "GuNi_RelativeCoords.csv"))
```

```
## Rows: 80 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (10): OID_, IN_FID, NEAR_FID, NEAR_DIST, NEAR_RANK, FROM_X, FROM_Y, NEAR...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
GuNi_Dist <- read_csv(here("EnteredData", "GuNi_Distance.csv"))
```

```
## Rows: 1560 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (10): OID_, IN_FID, NEAR_FID, NEAR_DIST, NEAR_RANK, FROM_X, FROM_Y, NEAR...
```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
GuNi_Locations <- read_csv(here("EnteredData", "GuNi_Location.csv"))
```

```
## Rows: 40 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (6): OID_, Comment, POINT_X, POINT_Y, POINT_Z, POINT_M
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Fixing Cover Names

```r
TeMi_Cover <- read_csv(here("EnteredData", "TeMi_Cover.csv"))
```

```
## Rows: 12 Columns: 196
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (1): Species
## dbl (190): TeMi.01f, TeMi.01n, TeMi.01e, TeMi.01s, TeMi.01w, TeMi.03f, TeMi....
## lgl   (5): TeMi.02f, TeMi.02n, TeMi.02e, TeMi.02s, TeMi.02w
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
TeMi_Cover<- merge(TeMi_Cover, Proper_Names, by.x = "Species", by.y = "Old_Species_Name")
rownames(TeMi_Cover)<- TeMi_Cover$New_Species_Name
TeMi_Cover<- TeMi_Cover[2:(length(TeMi_Cover)-1) ]

Zmud_Cover <- read_csv(here("EnteredData", "Zmud_Cover.csv"))
```

```
## Rows: 15 Columns: 216
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (1): Species
## dbl (215): Zmud.01f, Zmud.01n, Zmud.01e, Zmud.01s, Zmud.01w, Zmud.02f, Zmud....
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
Zmud_Cover<- merge(Zmud_Cover, Proper_Names, by.x = "Species", by.y = "Old_Species_Name")
rownames(Zmud_Cover)<- Zmud_Cover$New_Species_Name
Zmud_Cover<- Zmud_Cover[2:(length(Zmud_Cover)-1) ]

GuNi_Cover <- read_csv(here("EnteredData", "GuNi_Cover.csv"))
```

```
## Rows: 11 Columns: 201
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (1): Species
## dbl (155): GuNi.01f, GuNi.01n, GuNi.01e, GuNi.01s, GuNi.01w, GuNi.02f, GuNi....
## lgl  (45): GuNi.03f, GuNi.03n, GuNi.03e, GuNi.03s, GuNi.03w, GuNi.13f, GuNi....
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
GuNi_Cover<- merge(GuNi_Cover, Proper_Names, by.x = "Species", by.y = "Old_Species_Name")
rownames(GuNi_Cover)<- GuNi_Cover$New_Species_Name
GuNi_Cover<- GuNi_Cover[2:(length(GuNi_Cover)-1) ]
```

Create Relative Coords

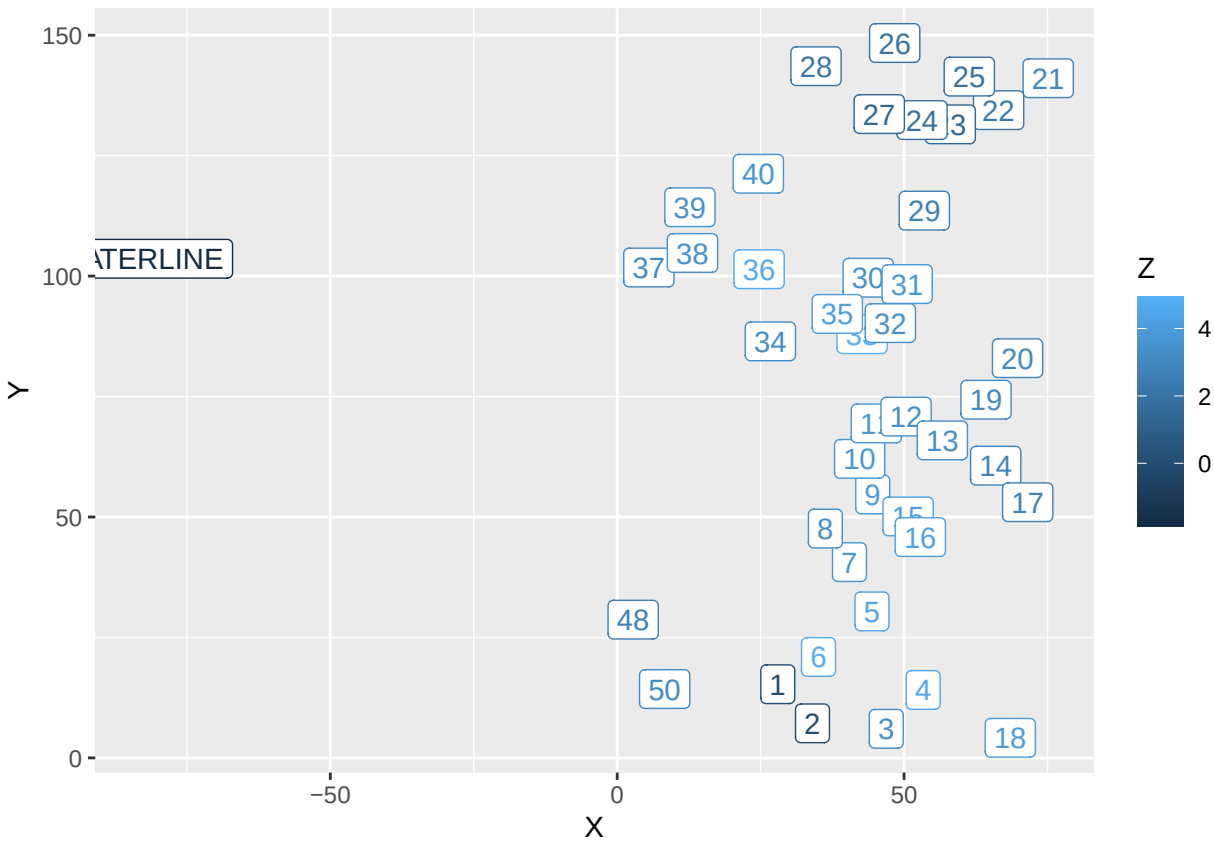
Here I am creating coordinates for each point at each site relative to the plot. Rather than being absolute, North South, these are positioned parrelel to the waterline. This will be used for drawing environmental comparisons along these axes which make more sense with stress.

```
#TeMi
TeMi_Subset_X <- subset(TeMi_Coords, IN_FID == 1) #Just the X
TeMi_Subset_Y <- subset(TeMi_Coords, IN_FID == 2) #Just the Y
TeMi_Sub<-merge(TeMi_Subset_X, TeMi_Subset_Y, by = "NEAR_FID") #recombine them
TeMi_RelCords<- data.frame(Plot= TeMi_Sub$NEAR_FID, X = TeMi_Sub$NEAR_DIST.x, Y = TeMi_Sub$NEAR_DIST.y,
TeMi_RelCords$Plot<- merge(TeMi_RelCords[1], TeMi_Locations[1:2], by.x = "Plot", by.y = "OID_")$Comment
TeMi_RelCords<- mutate(TeMi_RelCords, X = ifelse(TeMi_RelCords$Plot == "WATERLINE",-1*X, X)) #
TeMi_RelCords$Z <- TeMi_RelCords$Z - min(subset(TeMi_RelCords,Plot != "WATERLINE")$Z)

#Zmud
Zmud_Subset_X <- subset(Zmud_Coords, IN_FID == 2)
Zmud_Subset_Y <- subset(Zmud_Coords, IN_FID == 3)
Zmud_Sub<-merge(Zmud_Subset_X, Zmud_Subset_Y, by = "NEAR_FID")
Zmud_RelCords<- data.frame(Plot= Zmud_Sub$NEAR_FID, X = Zmud_Sub$NEAR_DIST.x, Y = Zmud_Sub$NEAR_DIST.y,
Zmud_RelCords$Plot<- merge(Zmud_RelCords[1], Zmud_Locations[1:2], by.x = "Plot", by.y = "OID_")$Comment
Zmud_RelCords<- mutate(Zmud_RelCords, X = ifelse(Zmud_RelCords$Plot == "WATERLINE",-1*X, X))
Zmud_RelCords$Z <- Zmud_RelCords$Z - min(subset(Zmud_RelCords,Plot != "WATERLINE")$Z)

#GuNi
GuNi_Subset_X <- subset(GuNi_Coords, IN_FID == 3)
GuNi_Subset_Y <- subset(GuNi_Coords, IN_FID == 2)
GuNi_Sub<-merge(GuNi_Subset_X, GuNi_Subset_Y, by = "NEAR_FID")
GuNi_RelCords<- data.frame(Plot= GuNi_Sub$NEAR_FID, X = GuNi_Sub$NEAR_DIST.x, Y = GuNi_Sub$NEAR_DIST.y,
GuNi_RelCords$Plot<- merge(GuNi_RelCords[1], GuNi_Locations[1:2], by.x = "Plot", by.y = "OID_")$Comment
GuNi_RelCords<- mutate(GuNi_RelCords, X = ifelse(GuNi_RelCords$Plot == "WATERLINE",-1*X, X))
GuNi_RelCords$Z <- GuNi_RelCords$Z - min(GuNi_RelCords$Z)
```

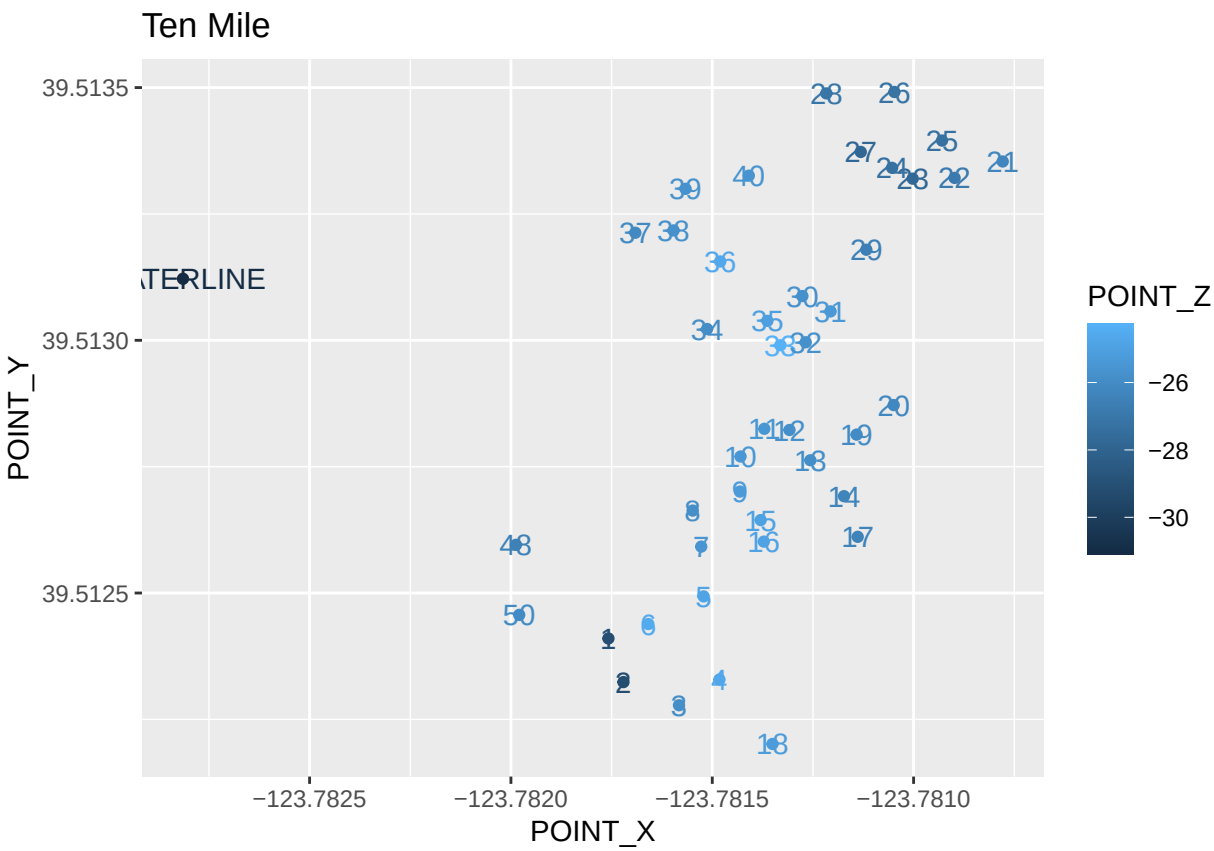
Checking Coords

Here I am making sure that the rotation worked properly.

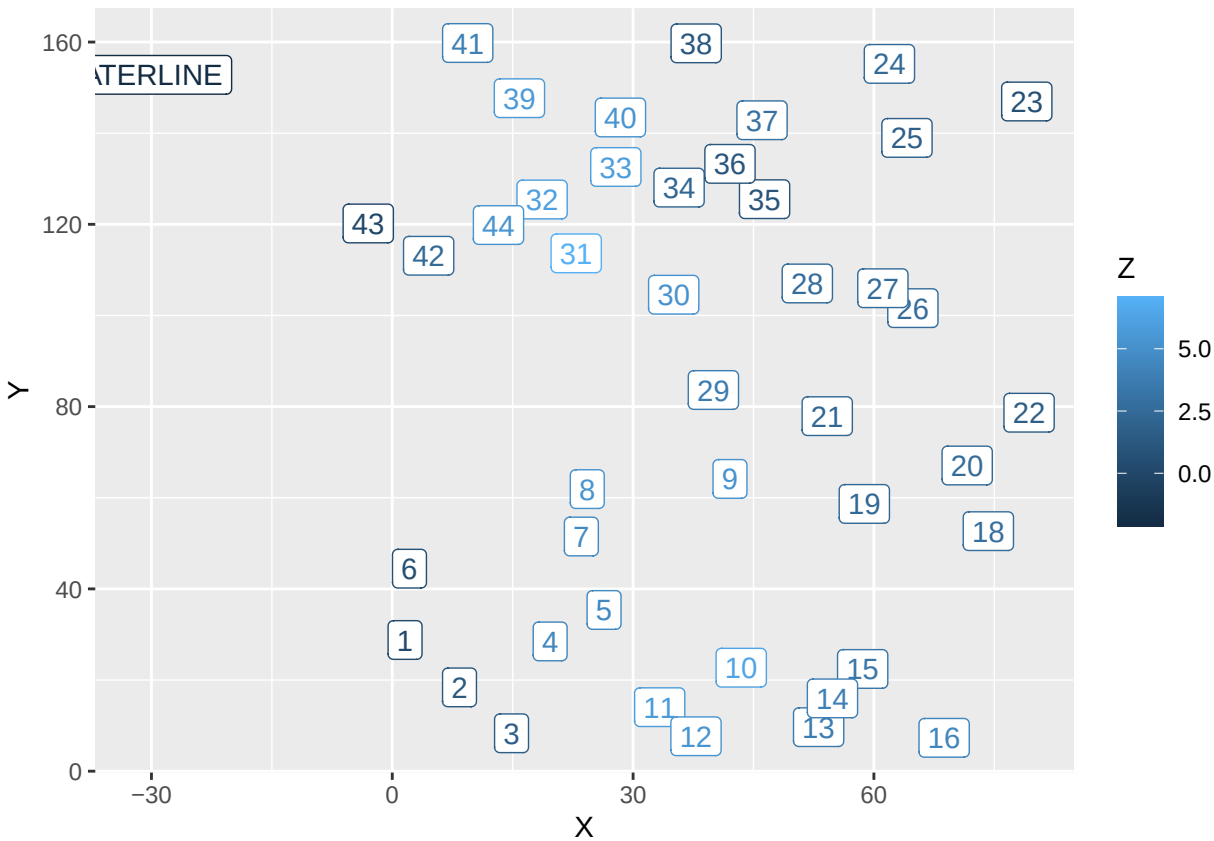
```
ggplot(aes(x = X, y = Y, label = Plot, color = Z), data = TeMi_RelCords)+
  geom_point()+
  geom_label()
```



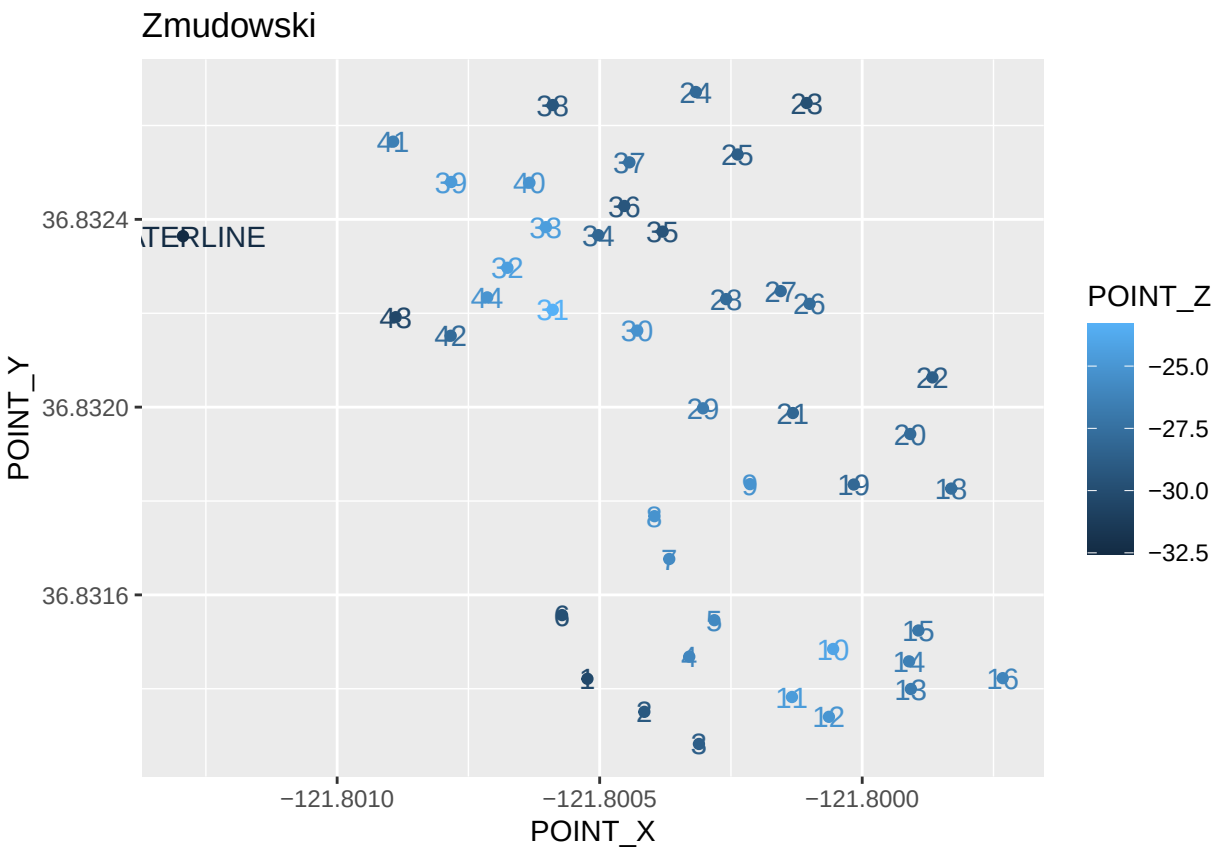

```
ggplot(aes(x = POINT_X, y = POINT_Y, label = Comment, color = POINT_Z), data = TeMi_Locations)+
  geom_point()+
  geom_text()+
  ggtitle("Ten Mile")
```



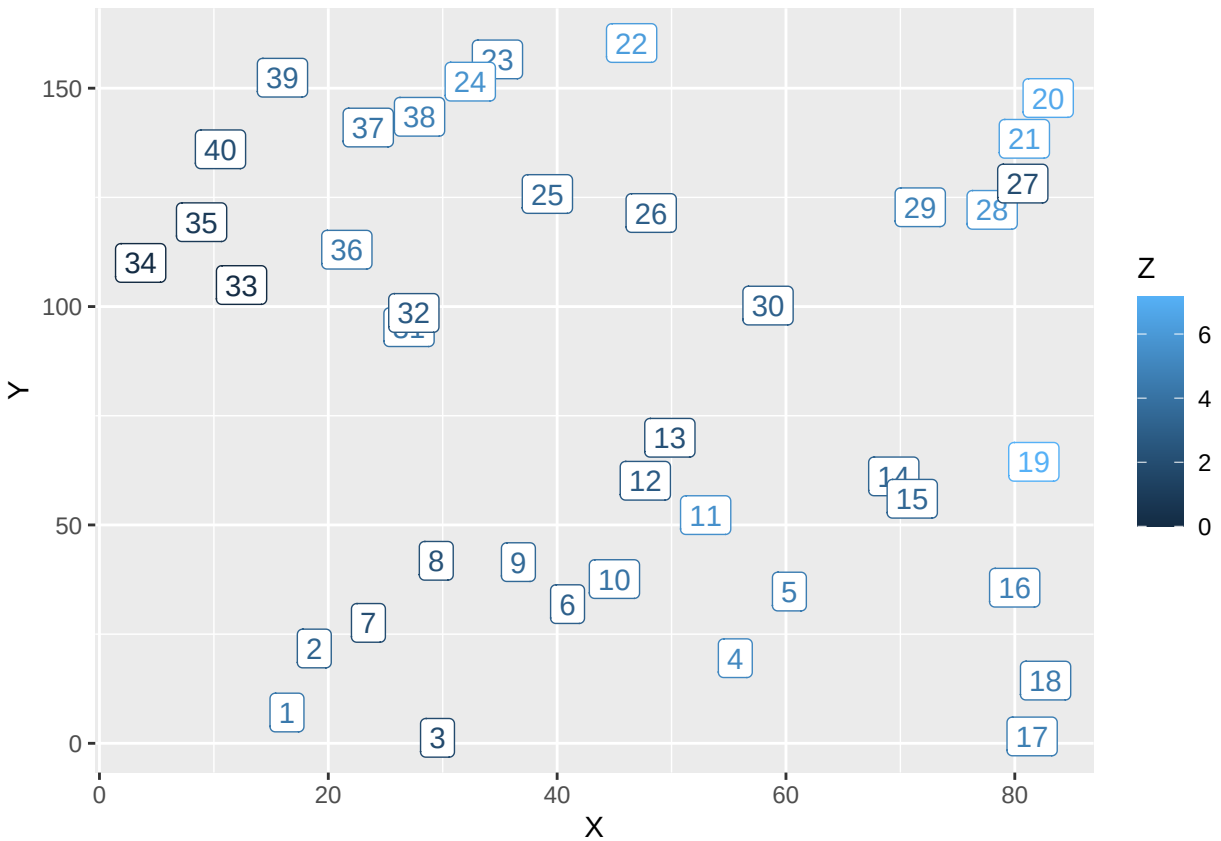

```r
ggplot(aes(x = X, y = Y, label = Plot, color = Z), data = Zmud_RelCords)+
  geom_point()+
  geom_label()
```



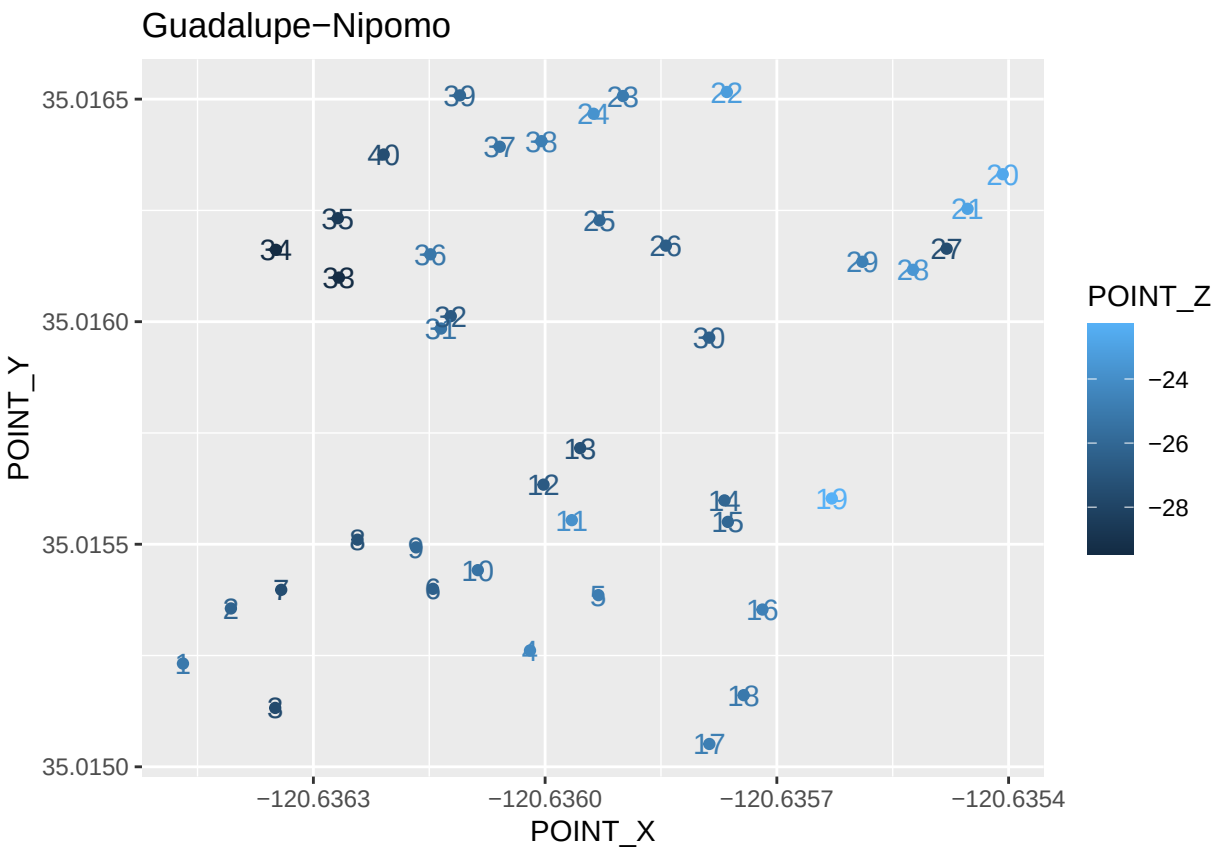

```
ggplot(aes(x = POINT_X, y = POINT_Y, label = Comment, color = POINT_Z), data = Zmud_Locations)+
  geom_point()+
  geom_text()+
  ggtitle("Zmudowski")
```




```
ggplot(aes(x = X, y = Y, label = Plot, color = Z), data = GuNi_RelCords)+
  geom_point()+
  geom_label()
```




```
ggplot(aes(x = POINT_X, y = POINT_Y, label = Comment, color = POINT_Z), data = GuNi_Locations)+
  geom_point()+
  geom_text()+
  ggtitle("Guadalupe-Nipomo")
```



Spanning Dataframes

Env Data

```
DuneEnv <- read_csv(here("EnteredData", "DuneEnvData.csv"))
```

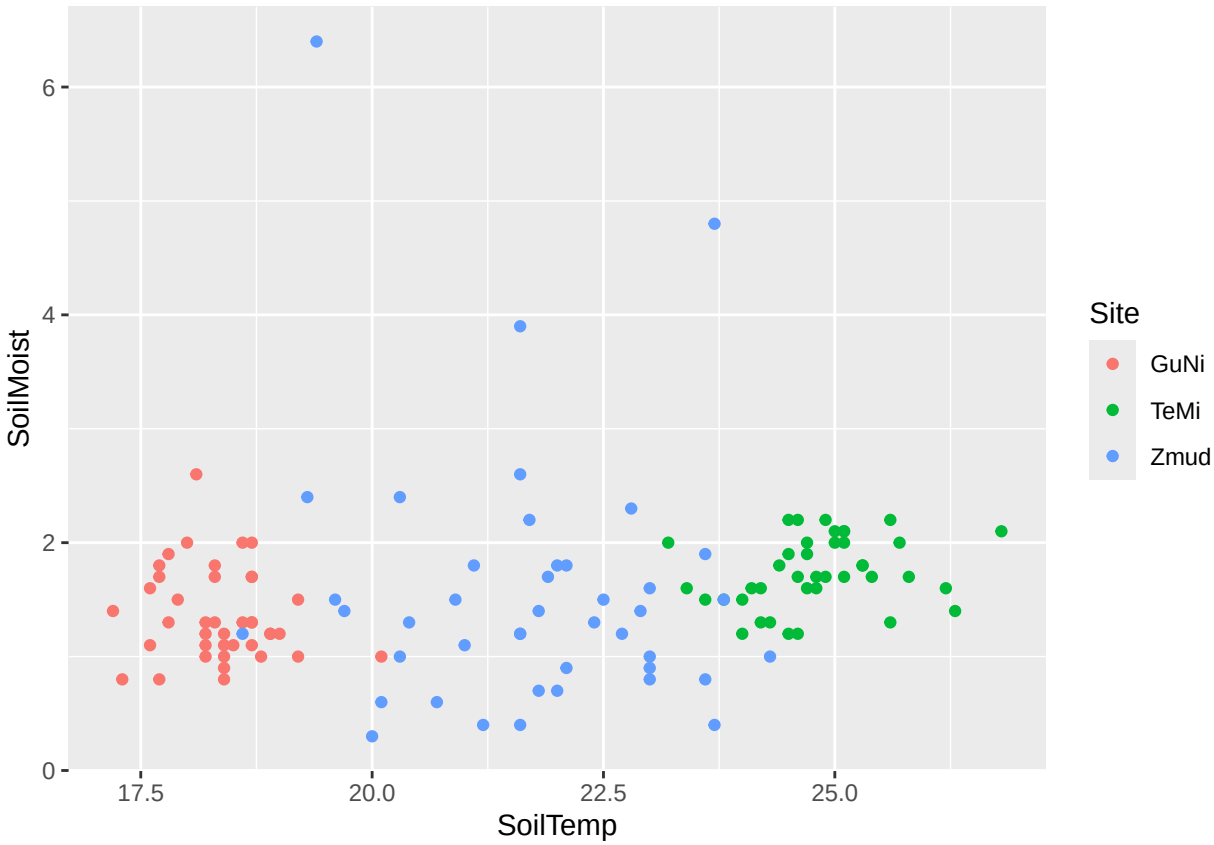
```
## Rows: 123 Columns: 8
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): Site, Aspect, Date, Collectors
## dbl (4): Plot, SoilTemp, SoilMoist, Slope
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
DuneSoil<-read_csv(here("EnteredData", "Dune_ph_salinity.csv"))
```

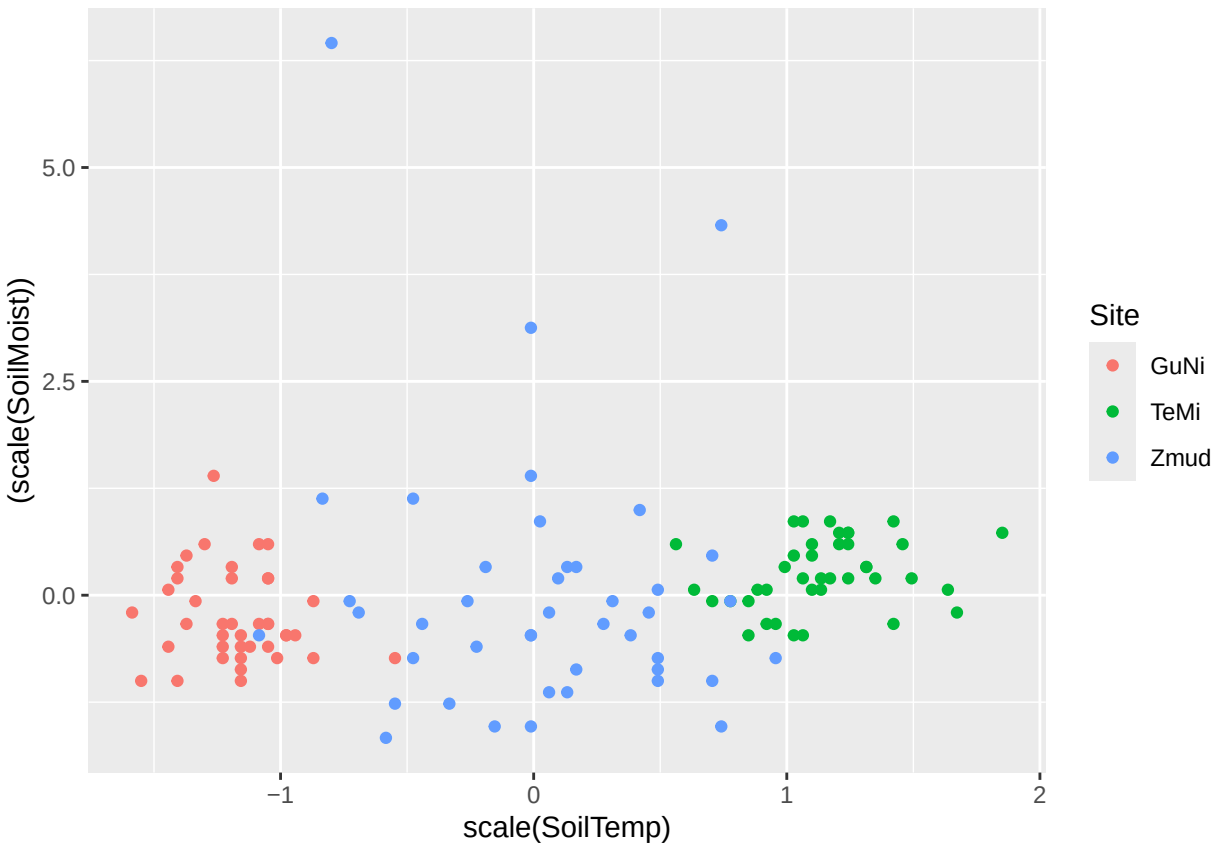
```
## Rows: 79 Columns: 7
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (2): Site, Notes
## dbl (5): Plot, dry_weight, pH, cond_us.cm, salinity_ppt
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
DuneEnvData<- merge(DuneEnv, DuneSoil, all = TRUE )
```

```r
ggplot(aes(x= SoilTemp, y = SoilMoist, color = Site), data = DuneEnvData)+
  geom_point() #okay so site super mattered which makes sense for temp. I'll scale it
```

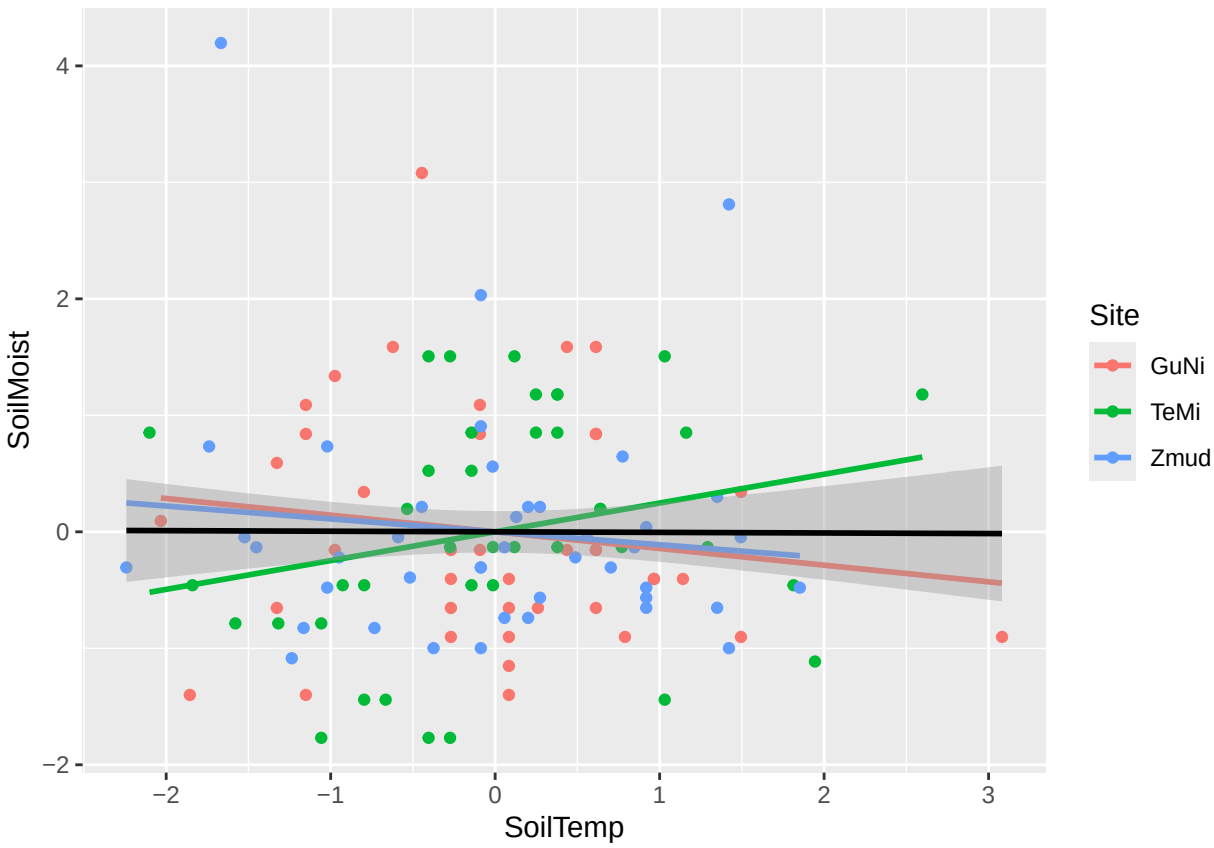



```r
ggplot(aes(x= scale(SoilTemp), y = (scale(SoilMoist)), color = Site), data = DuneEnvData)+
  geom_point() #Lol. I need to scale Soil Temp by the site! Duh! I should actually do that for Soil Moi
```




```
#DuneEnv_scaled <- DuneEnvData %>% group_by(Site) %>% mutate(SoilTemp = scale(SoilTemp), SoilMoist=scal

DuneEnv_scaled<-plyr::ddply(DuneEnvData, "Site", transform, SoilMoist = scale(SoilMoist))

DuneEnv_scaled<-plyr::ddply(DuneEnv_scaled, "Site", transform, SoilTemp = scale(SoilTemp))

DuneEnv_scaled$SoilTemp <- DuneEnv_scaled$SoilTemp[,1]

#Temp v moist
ggplot(aes(x= SoilTemp, y = SoilMoist), data = DuneEnv_scaled)+
  geom_point(aes(color = Site)) +
  geom_smooth(method = "lm", aes(color = Site), fill = NA)+
  geom_smooth(method = "lm", color = "black")
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```

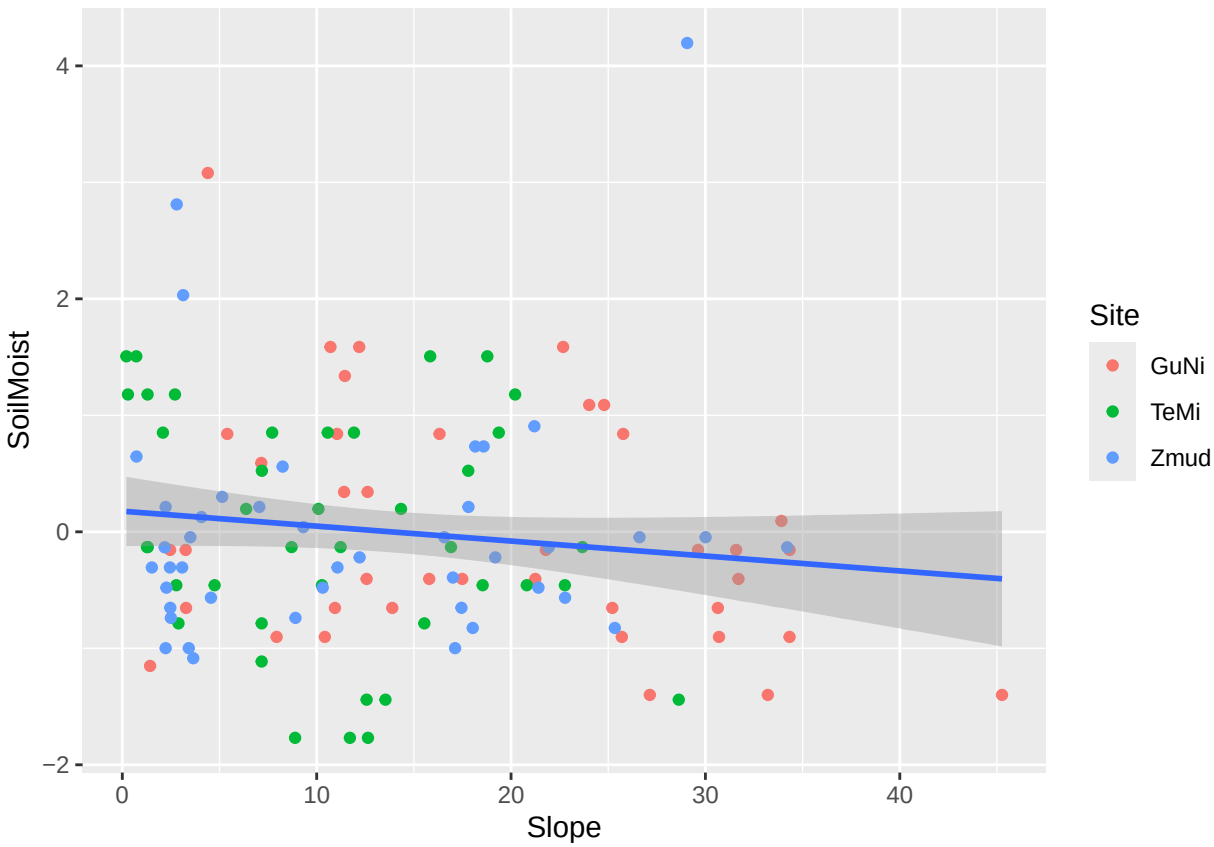



```r
cor.test(x = DuneEnv_scaled$SoilMoist, y = DuneEnv_scaled$SoilTemp, method = "pearson") # temp and mosi
```

```
##
##  Pearson's product-moment correlation
##
## data:  DuneEnv_scaled$SoilMoist and DuneEnv_scaled$SoilTemp
## t = -0.054767, df = 121, p-value = 0.9564
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.1818528  0.1722074
## sample estimates:
##          cor
## -0.004978767
```

```r
# Slope v SoilMoist
ggplot(aes(x= Slope, y = SoilMoist), data = DuneEnv_scaled)+
  geom_point(aes(color = Site)) +
  geom_smooth(method = "lm")# No obvious pattern
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

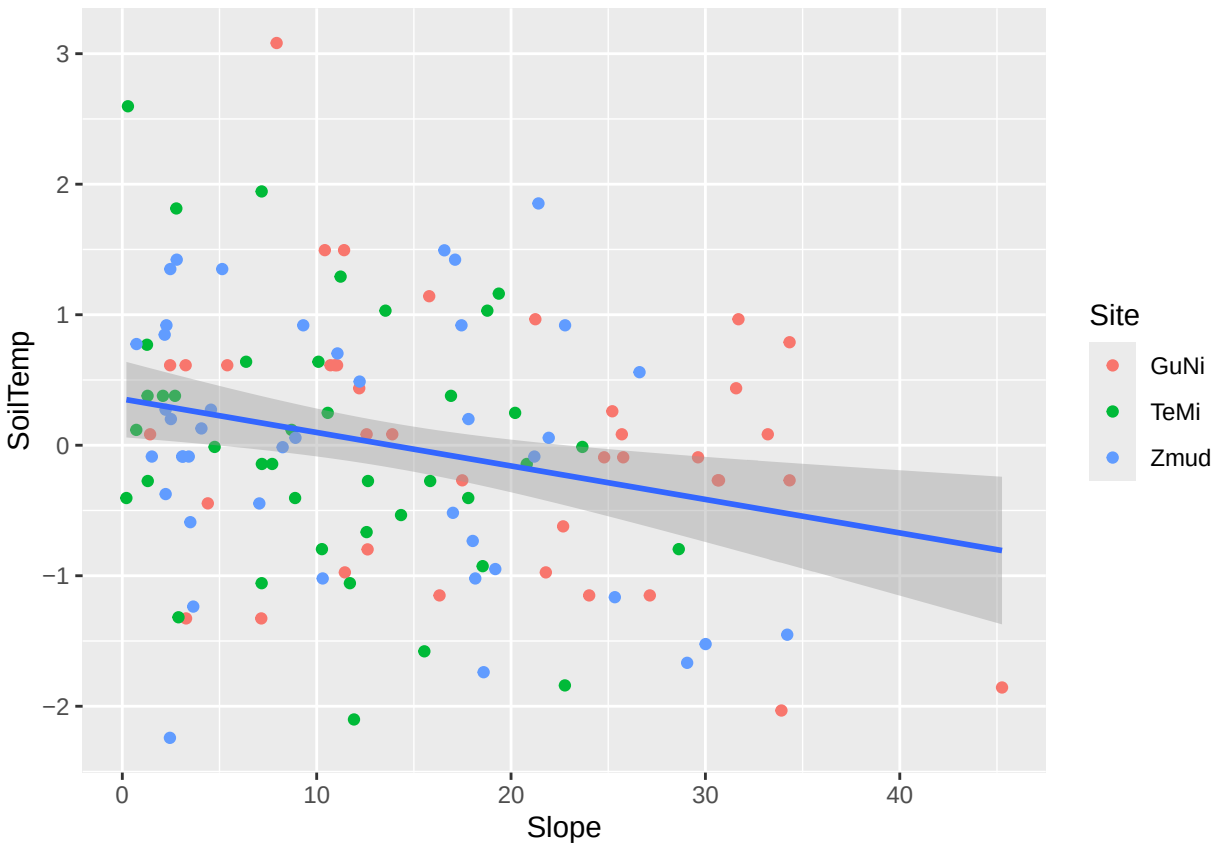



```r
cor.test(x = DuneEnv_scaled$Slope, y = DuneEnv_scaled$SoilMoist, method = "pearson") # Nutthin
```

```
##
##  Pearson's product-moment correlation
##
## data:  DuneEnv_scaled$Slope and DuneEnv_scaled$SoilMoist
## t = -1.4435, df = 121, p-value = 0.1515
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.30022885  0.04803074
## sample estimates:
##        cor
## -0.1301099
```

```r
#Slope v Temp ***
ggplot(aes(x= Slope , y =SoilTemp), data = DuneEnv_scaled)+
  geom_point(aes(color = Site)) +
  geom_smooth(method = "lm")
```

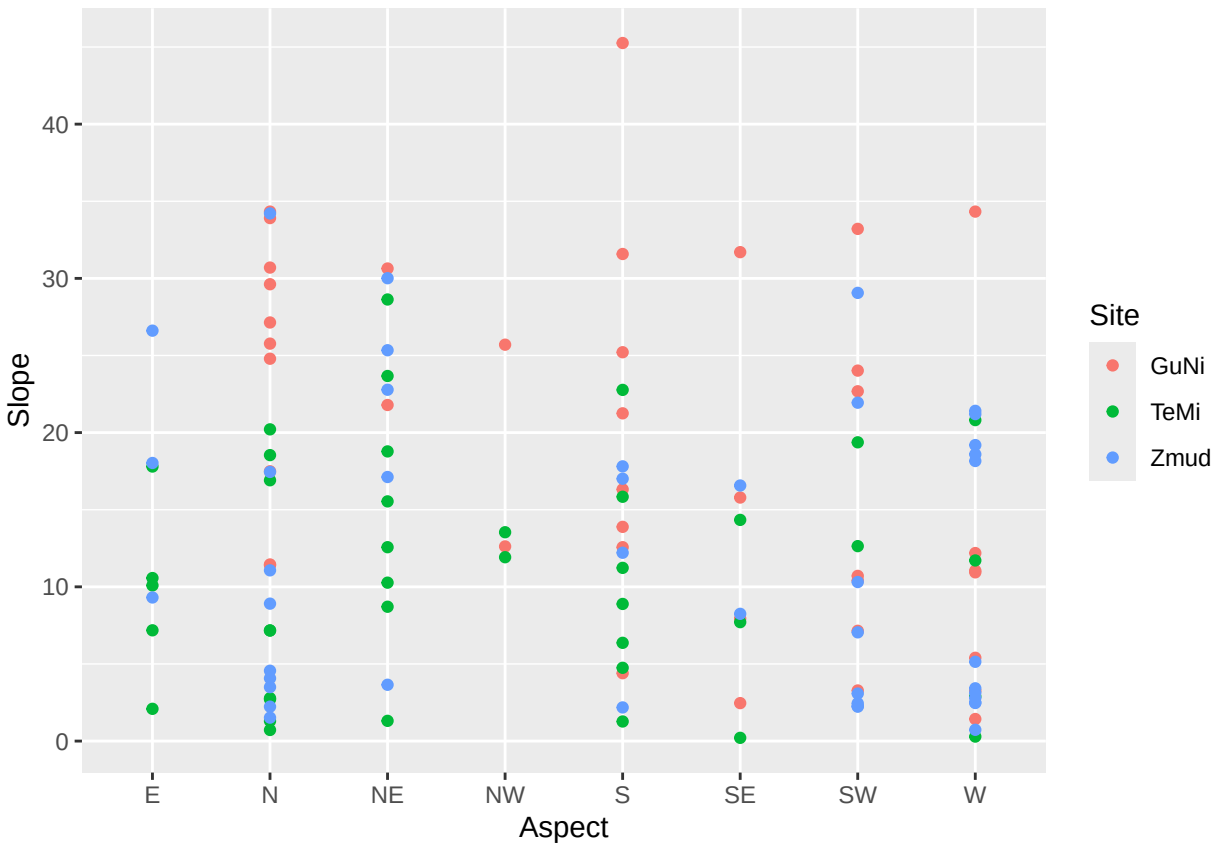
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
cor.test(x = DuneEnv_scaled$Slope, y = DuneEnv_scaled$SoilTemp, method = "pearson") # Topsoil, Slope ex
```

```
##
##  Pearson's product-moment correlation
##
## data:  DuneEnv_scaled$Slope and DuneEnv_scaled$SoilTemp
## t = -2.9647, df = 121, p-value = 0.003651
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.41800607 -0.08721253
## sample estimates:
##       cor
## -0.260229
```

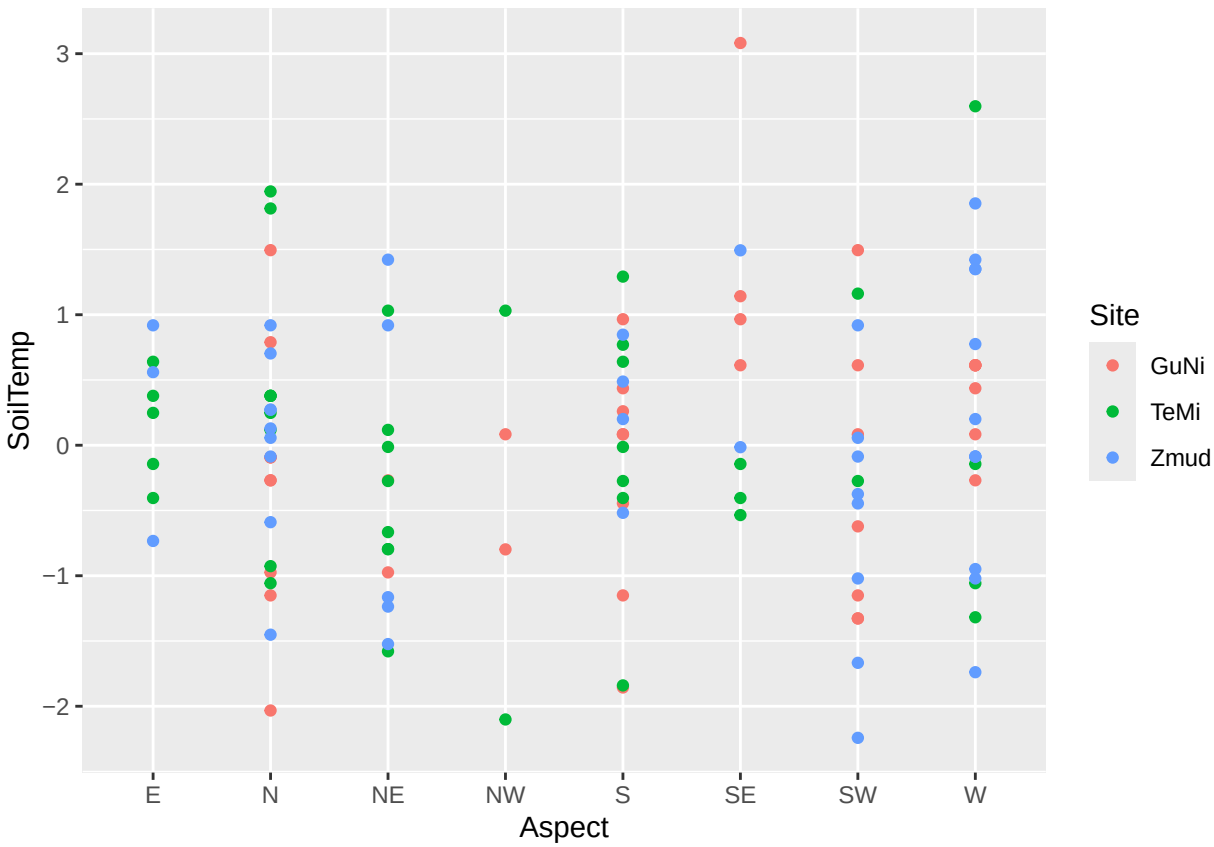
```
#aspect slope
ggplot(aes(x= Aspect, y = Slope), data = DuneEnv_scaled)+
  geom_point(aes(color = Site)) # No obvious pattern
```




```r
summary(aov(Slope ~ Aspect, data = DuneEnv_scaled)) #nuthin
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## Aspect        7    725   103.6   1.025  0.417
## Residuals   115  11620   101.0
```

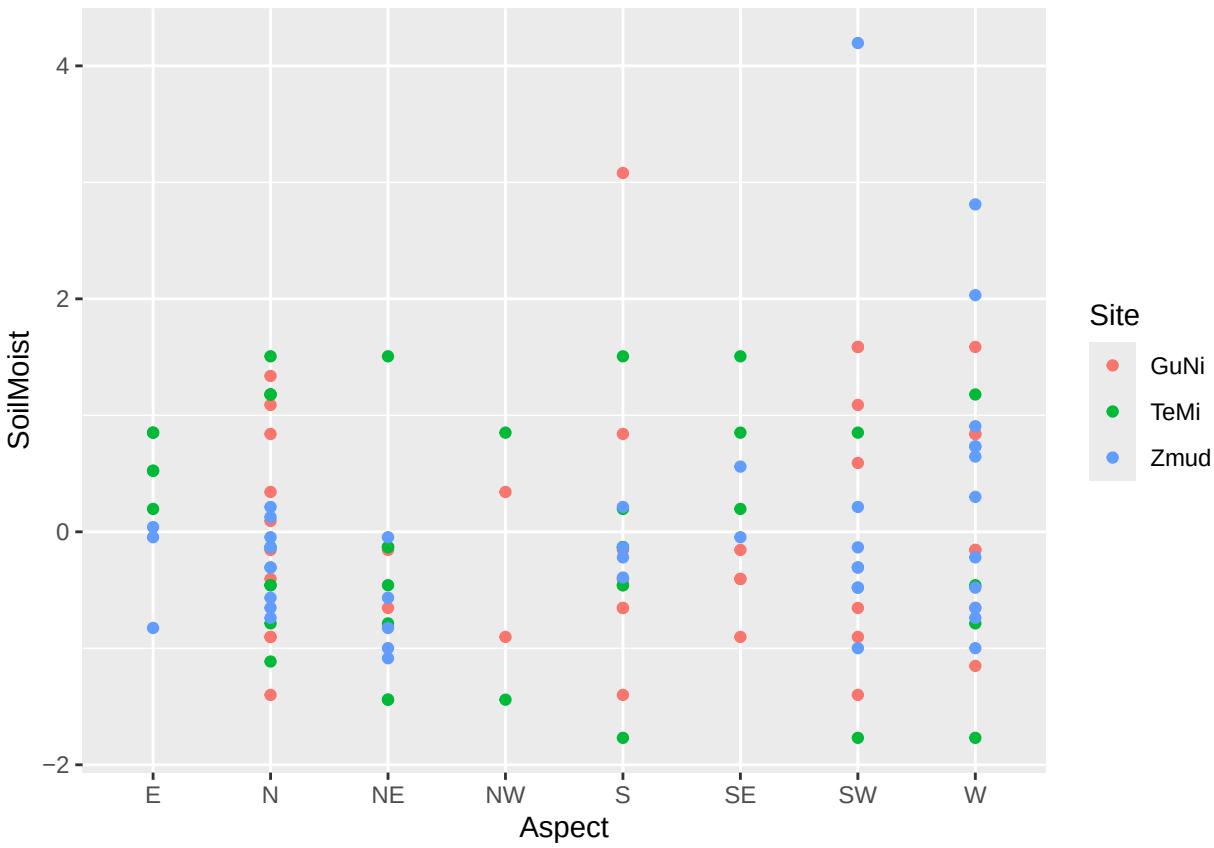
```r
#aspect temp
ggplot(aes(x= Aspect, y = SoilTemp), data = DuneEnv_scaled)+
  geom_point(aes(color = Site)) # No obvious pattern
```




```r
summary(aov(SoilTemp ~ Aspect, data = DuneEnv_scaled)) #nuthin
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## Aspect        7  11.32   1.617   1.711  0.113
## Residuals   115 108.68   0.945
```

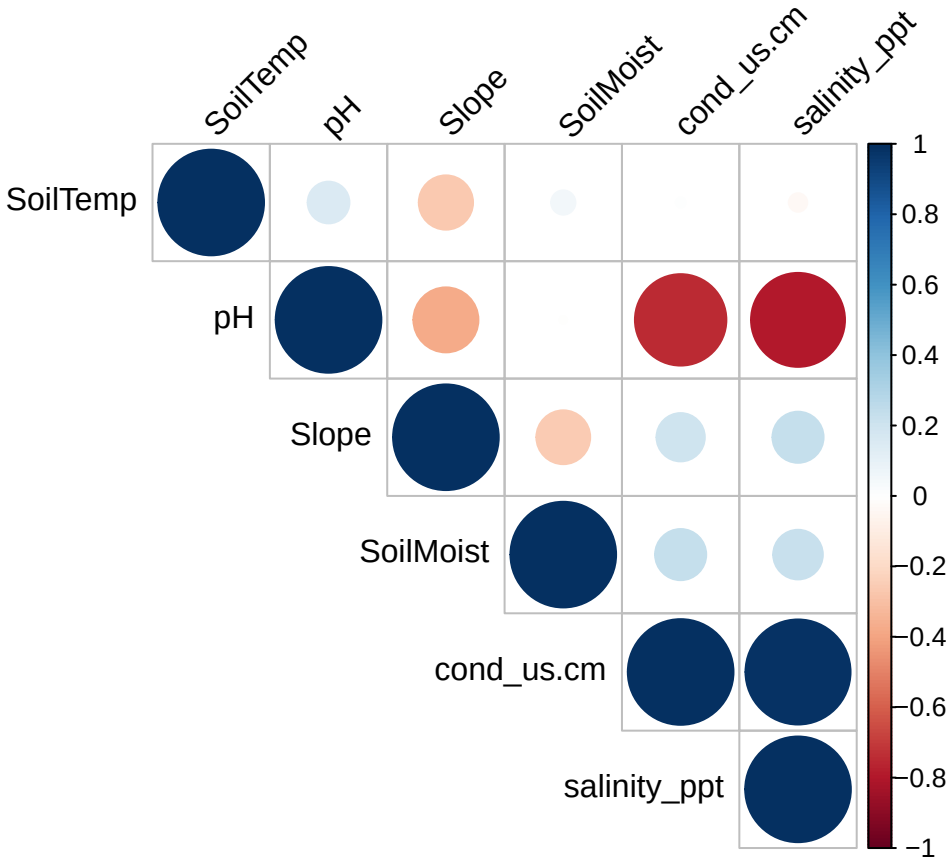
```r
#aspect Moist
ggplot(aes(x= Aspect, y = SoilMoist), data = DuneEnv_scaled)+
  geom_point(aes(color = Site)) # No obvious pattern
```




```r
summary(aov(SoilMoist ~ Aspect, data = DuneEnv_scaled)) #nuthin
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## Aspect        7   6.04  0.8623    0.87  0.532
## Residuals   115 113.96  0.9910
```

```r
DuneEnv_scaled %>%
  select(-c(Site, Plot, Aspect, Date, Collectors, dry_weight, Notes))%>%
  as.matrix()%>%
  cor(use =  "complete")%>%
  corrplot( type = "upper", order = "hclust",  tl.col = "black", tl.srt = 45)
```



Phenology

```
Pheno <-read_csv(here("EnteredData", "DunePhenoTrait.csv"))
```

```
## Rows: 404 Columns: 9
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): Site, Species, Phenology
## dbl (6): Plot, LeafLength, LeafWidth, LeafThickness, Height, BasalDiam
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

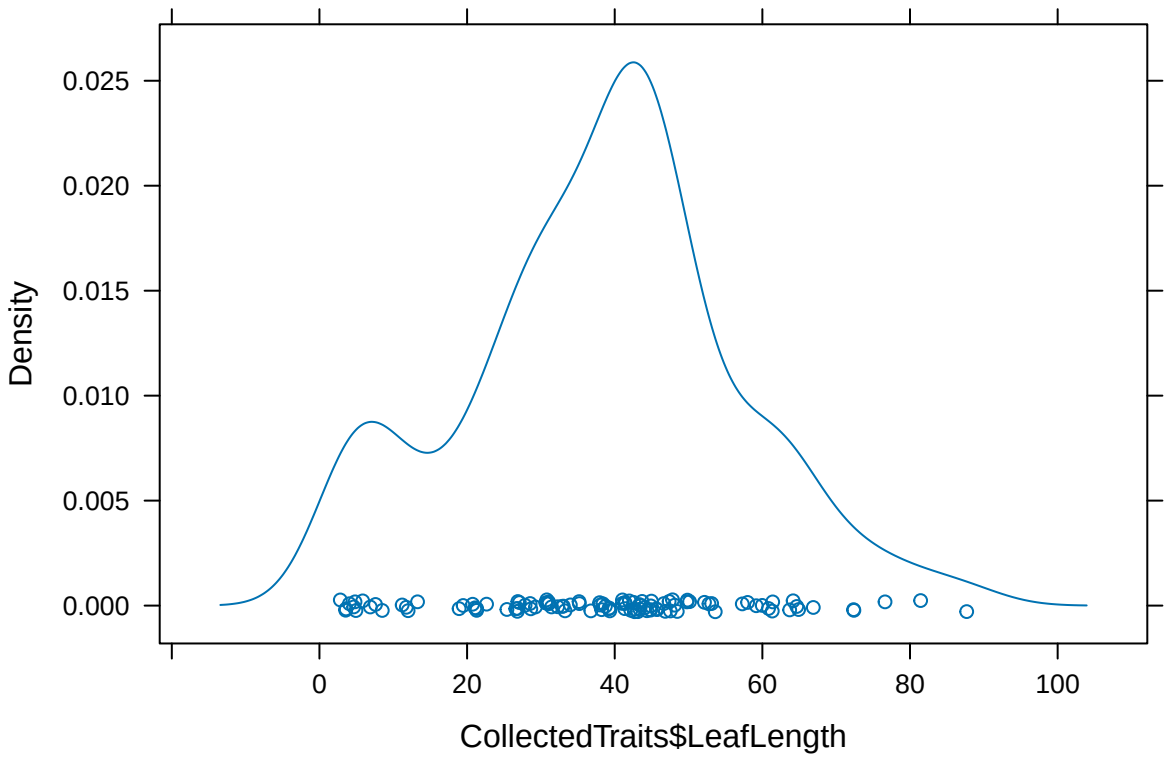
```
head(Pheno)
```

```
## # A tibble: 6 x 9
##   Site   Plot Species        Phenology LeafLength LeafWidth LeafThickness Height
##   <chr> <dbl> <chr>          <chr>          <dbl>     <dbl>         <dbl>  <dbl>
## 1 Zmud      1 Sea Rocket     fs             30.8       7.47          2.03   89.3
## 2 Zmud      2 Yellow Sand V~ v              64.2      48.2           3.69   47.7
## 3 Zmud      2 Morning glory  v              21.1      12.4           1.84   21.8
## 4 Zmud      2 Beach primrose fs              8.51      5.39          0.69   39.3
## 5 Zmud      2 Sea Rocket     s              26.8       5.92          2.64   33.6
## 6 Zmud      2 Carpobrotus e~ v              38.4       8.91          7.58   57.6
## # i 1 more variable: BasalDiam <dbl>
```

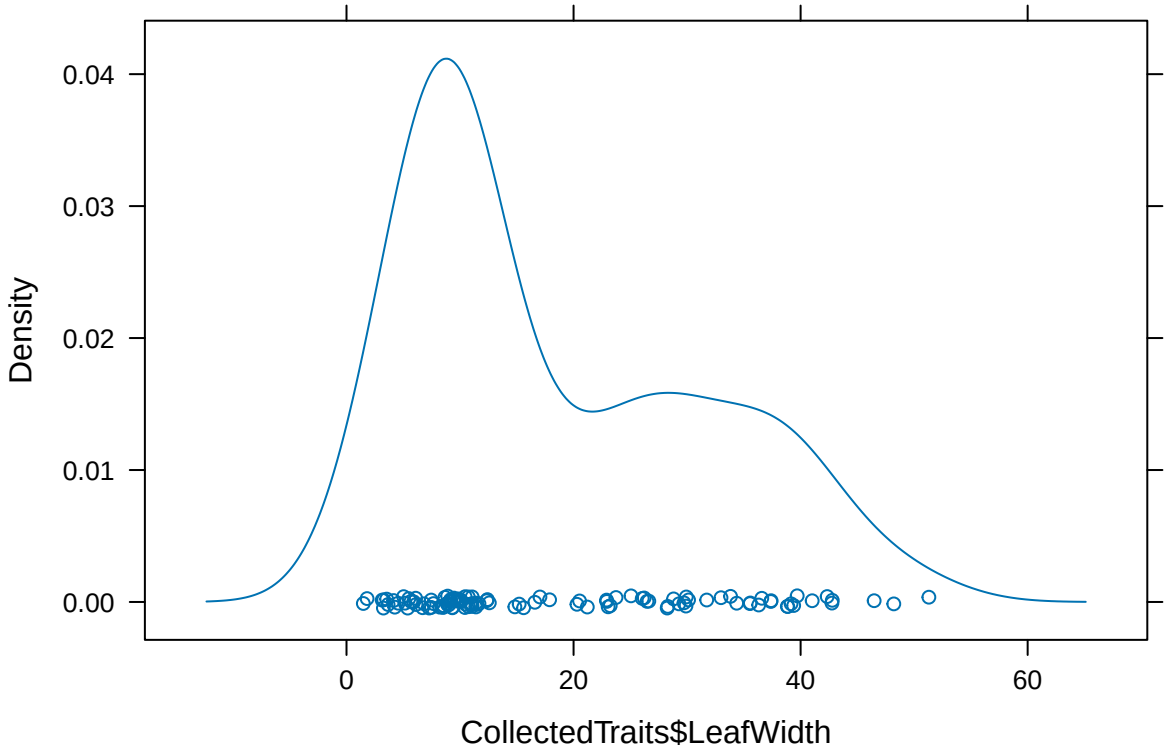
```r
CollectedTraits <- subset(Pheno,  is.na(LeafLength) == FALSE)


SuccTraits <- rbind(subset(CollectedTraits, Species == "Yellow Sand Verbena"), subset(CollectedTraits, S
```

Checking Graphs

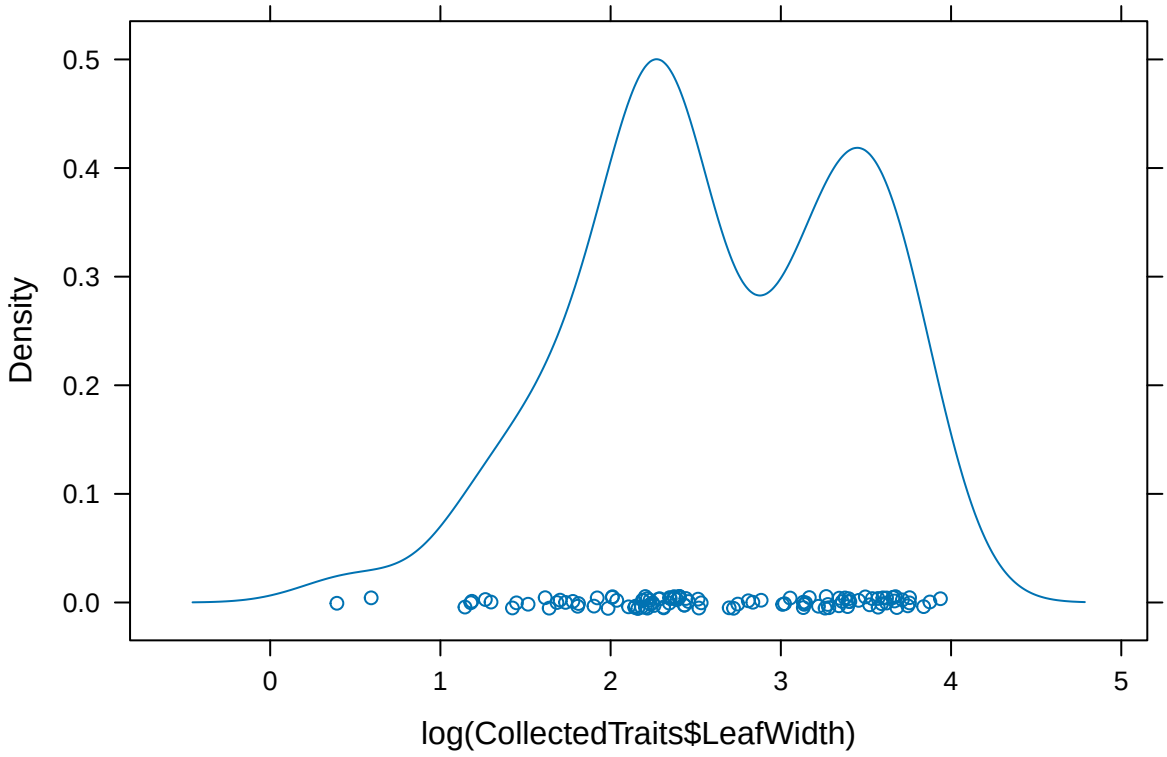
```r
densityplot(CollectedTraits$LeafLength) #Normal
```



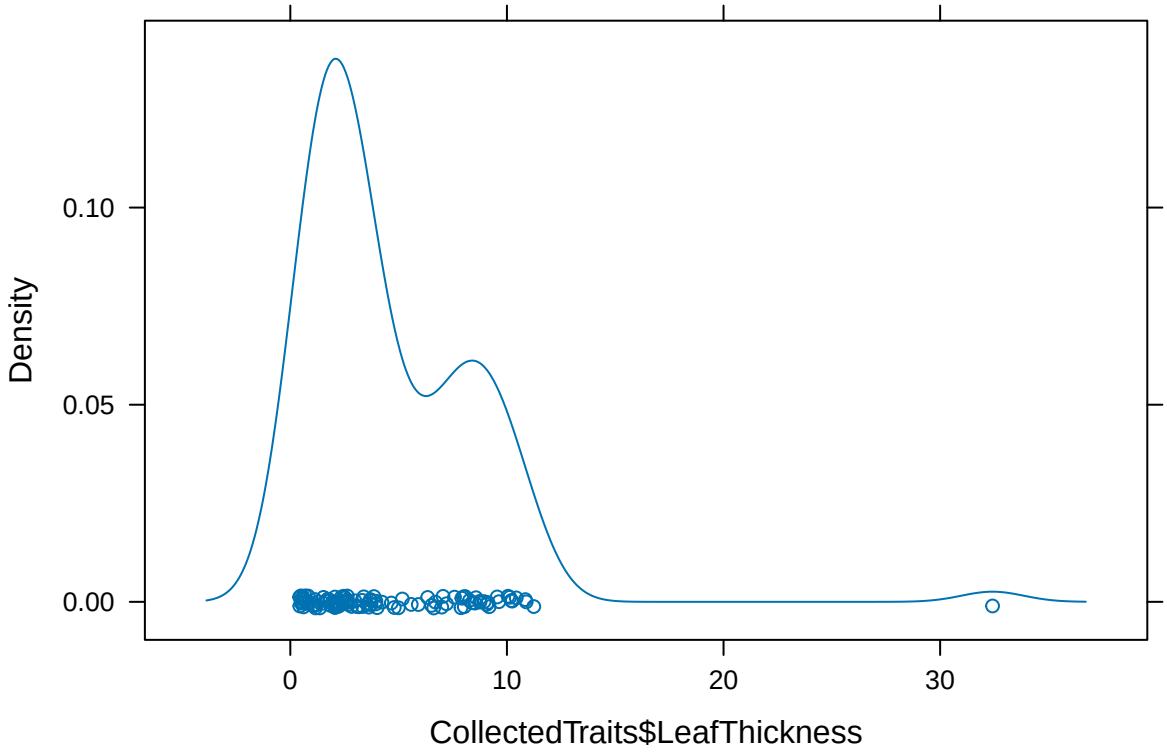

```r
densityplot(CollectedTraits$LeafWidth) #might need Log
```



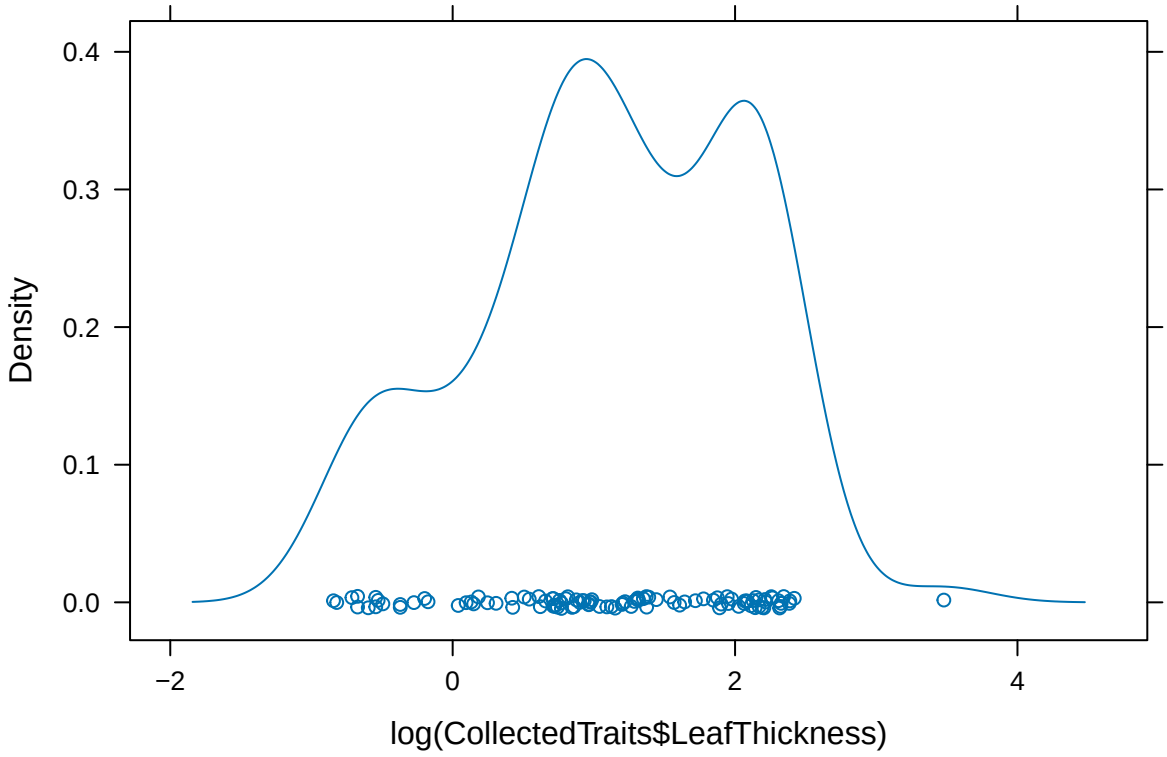

```r
densityplot(log(CollectedTraits$LeafWidth)) #not really better
```



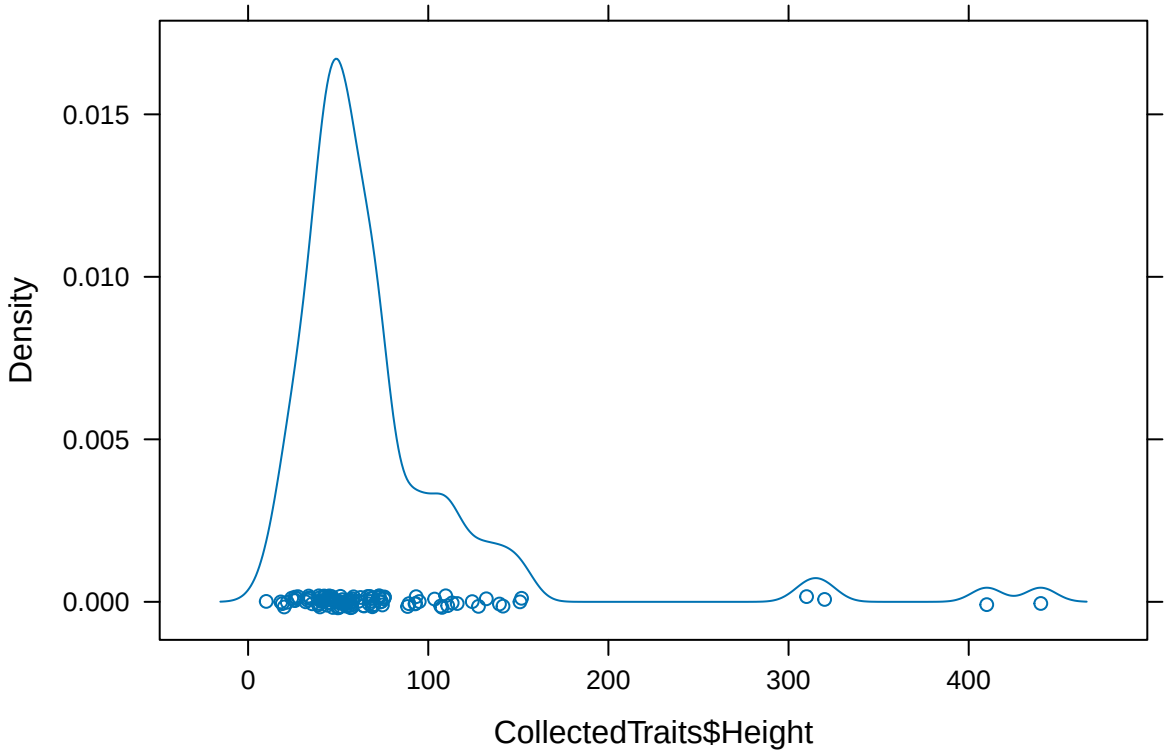

```r
densityplot(CollectedTraits$LeafThickness) #typo and log transform
```



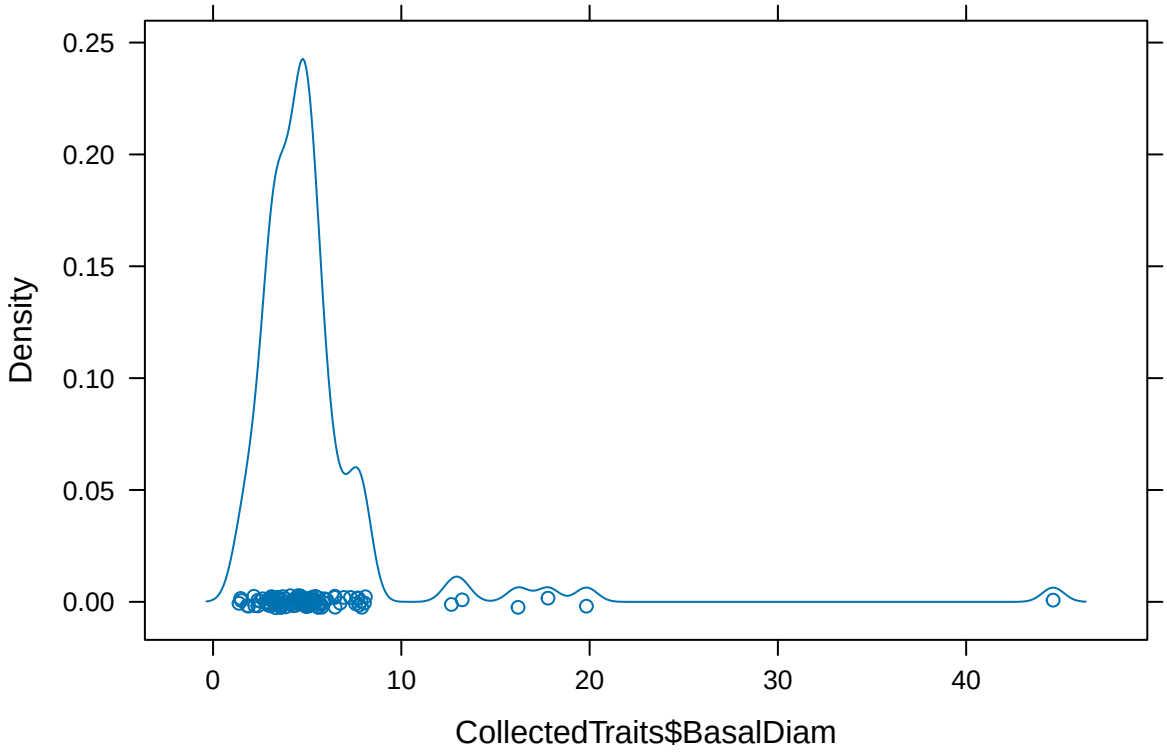

```
densityplot(log(CollectedTraits$LeafThickness))
```




```
densityplot(CollectedTraits$Height) #
```

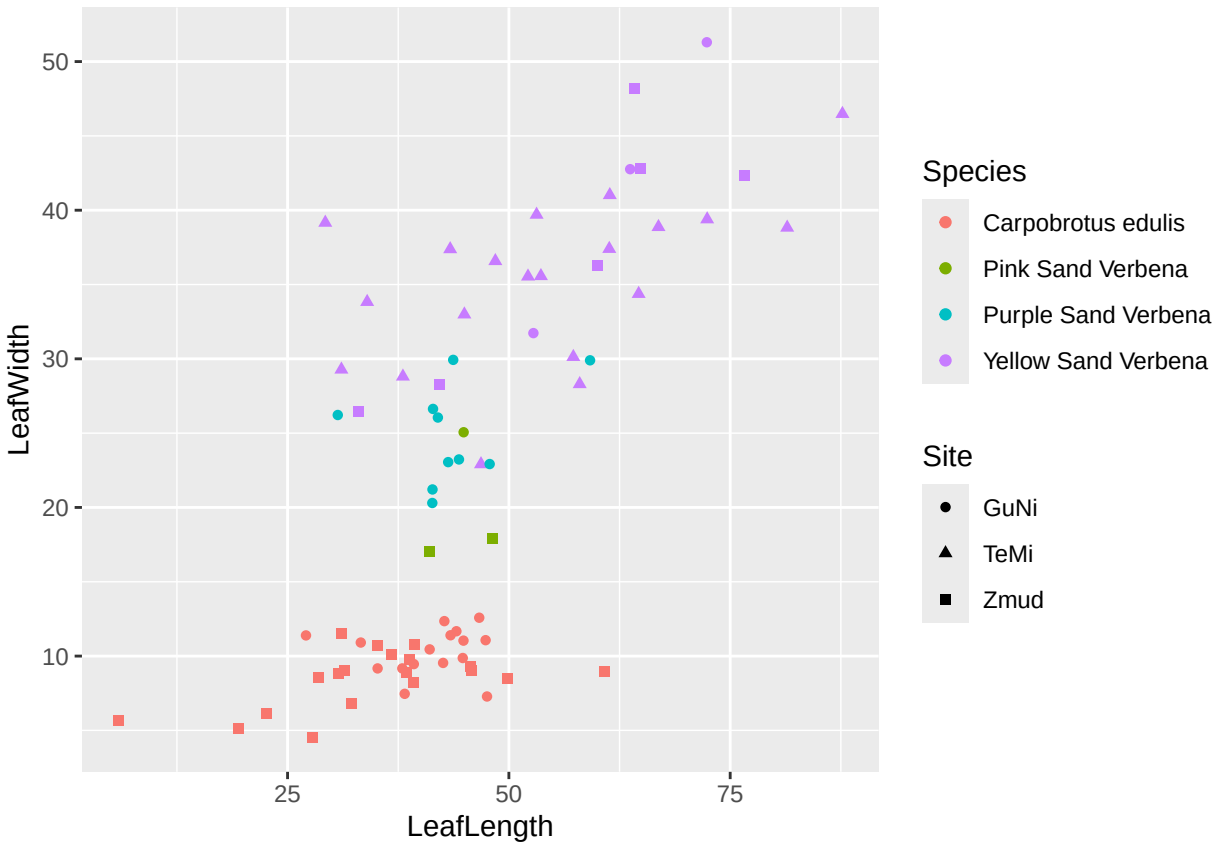



```r
densityplot(CollectedTraits$BasalDiam)
```

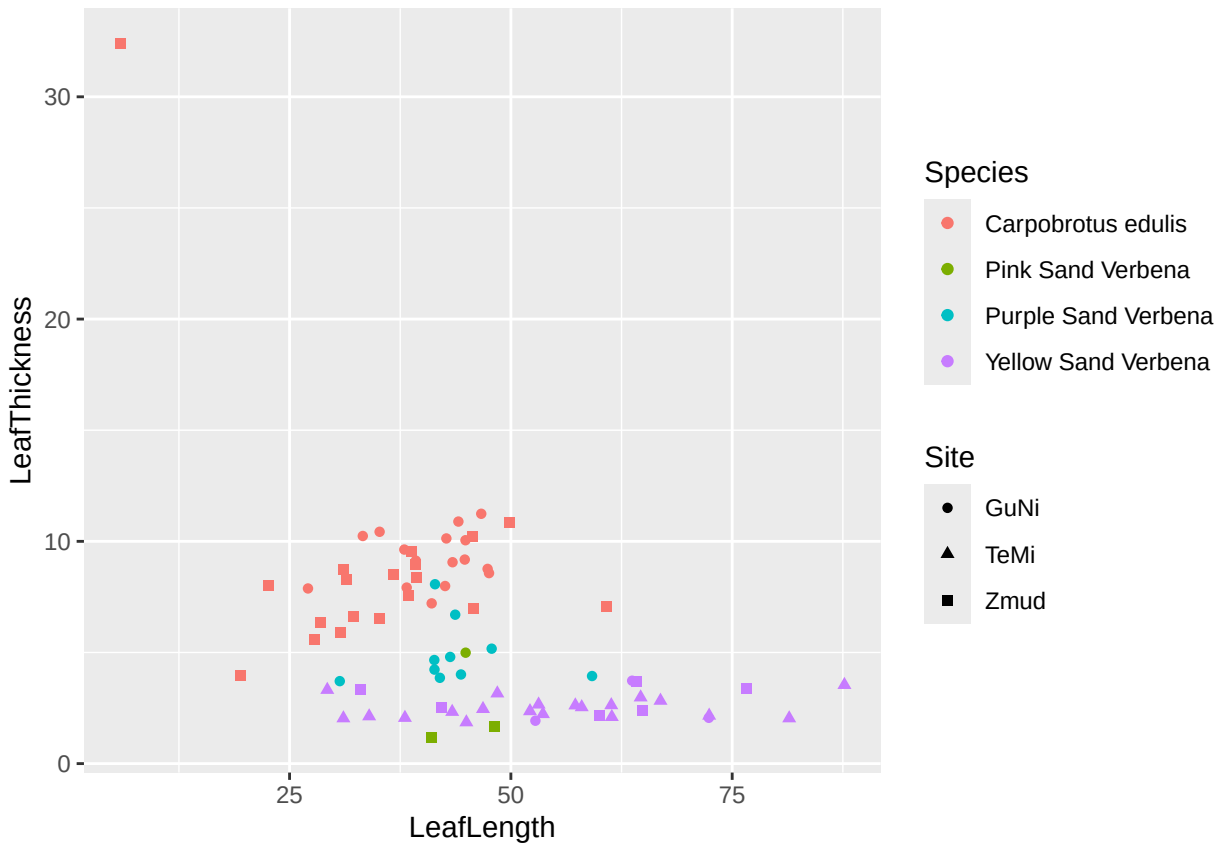



```r
#subset(CollectedTraits, LeafTraits$LeafThickness >20) #Thats a typo
#subset(CollectedTraits, LeafTraits$Height >205) #Okay, maybe legit
#subset(CollectedTraits, LeafTraits$BasalDiam >25) #Just a chonky boi but worth double checking
```

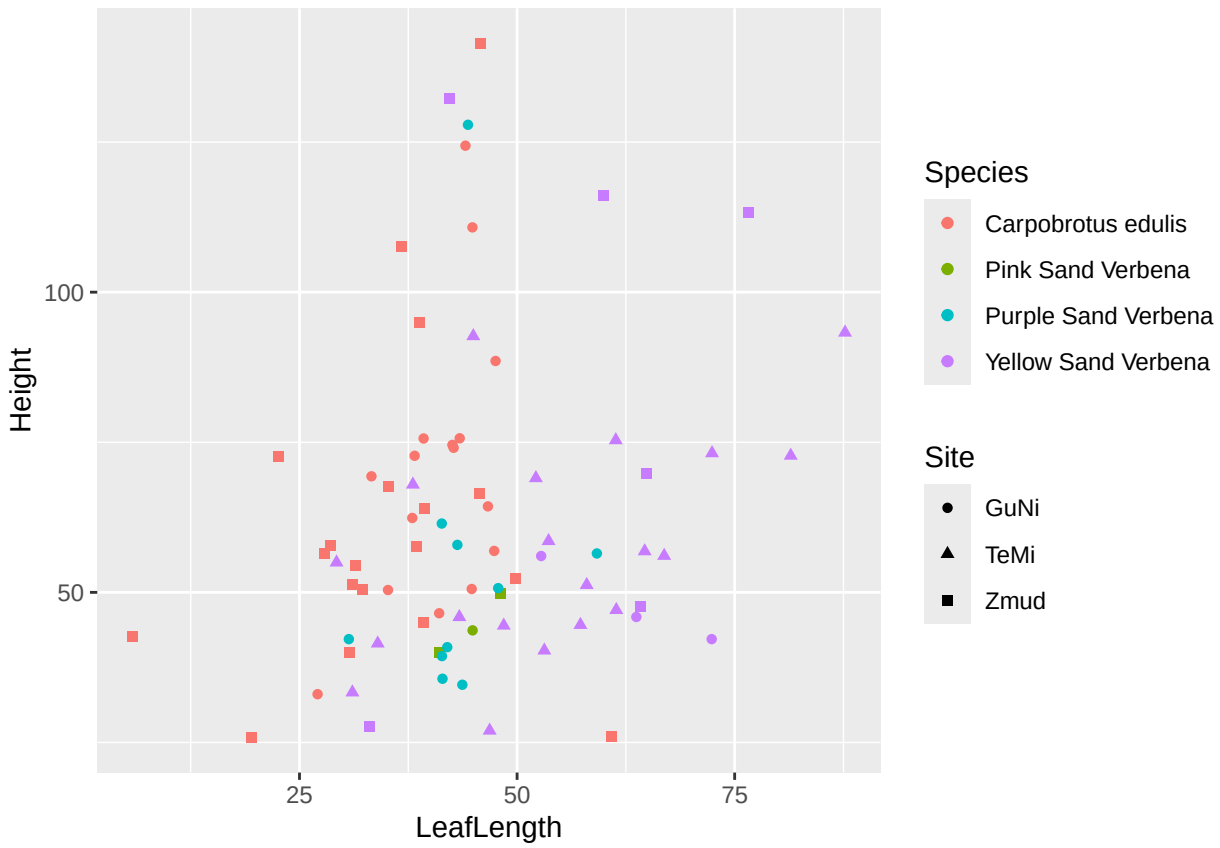
```r
ggplot(aes(x = LeafLength, y = LeafWidth, color = Species, shape = Site), data = SuccTraits)+
  geom_point()
```



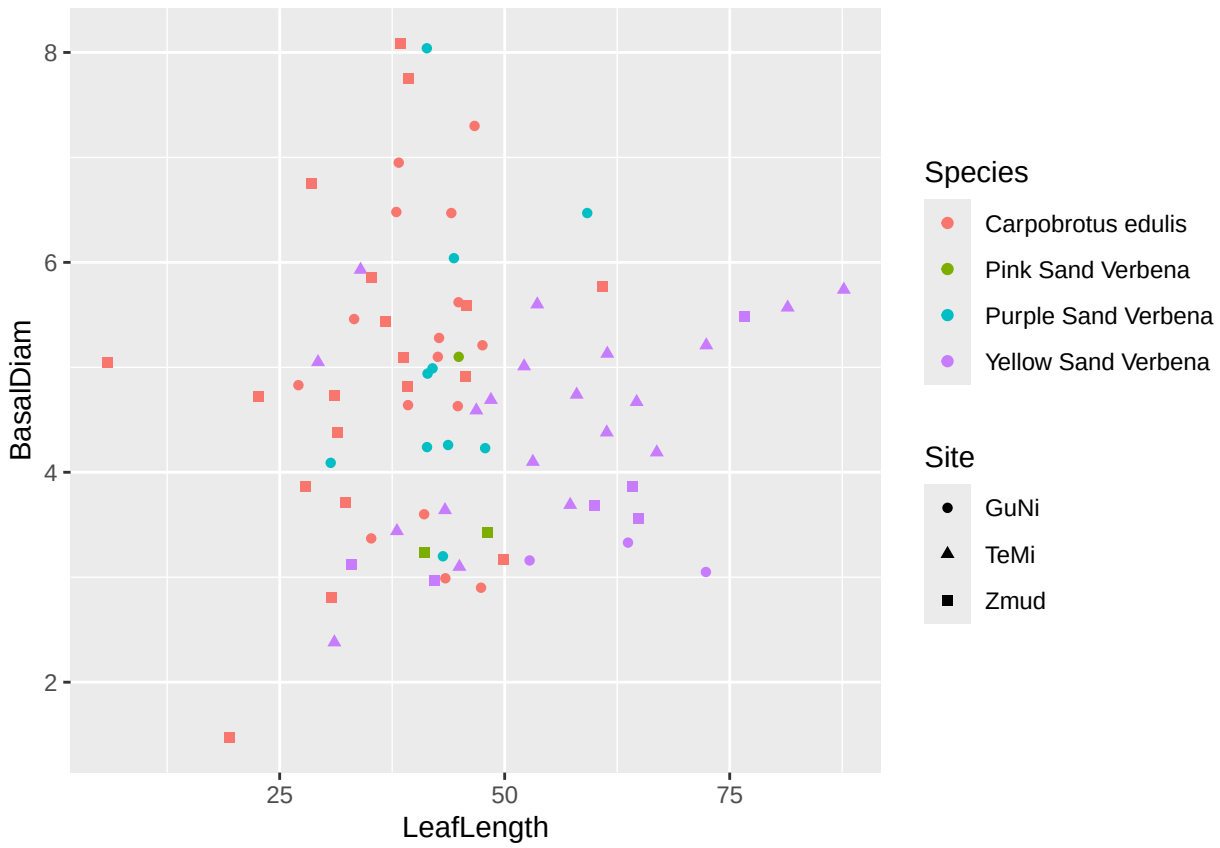

```r
ggplot(aes(x = LeafLength, y = LeafThickness, color = Species, shape = Site), data = SuccTraits)+
  geom_point()
```



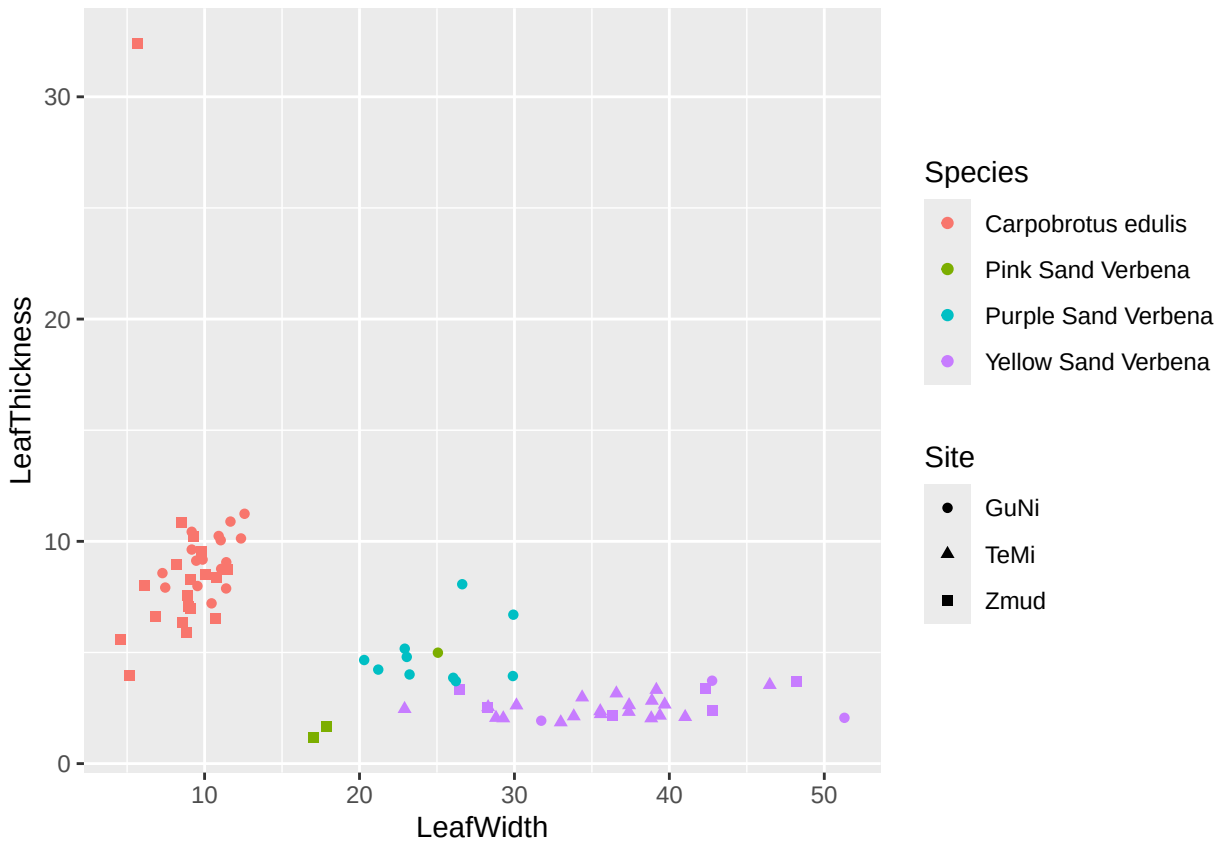

```
ggplot(aes(x = LeafLength, y = Height, color = Species, shape = Site), data = SuccTraits)+
  geom_point()
```



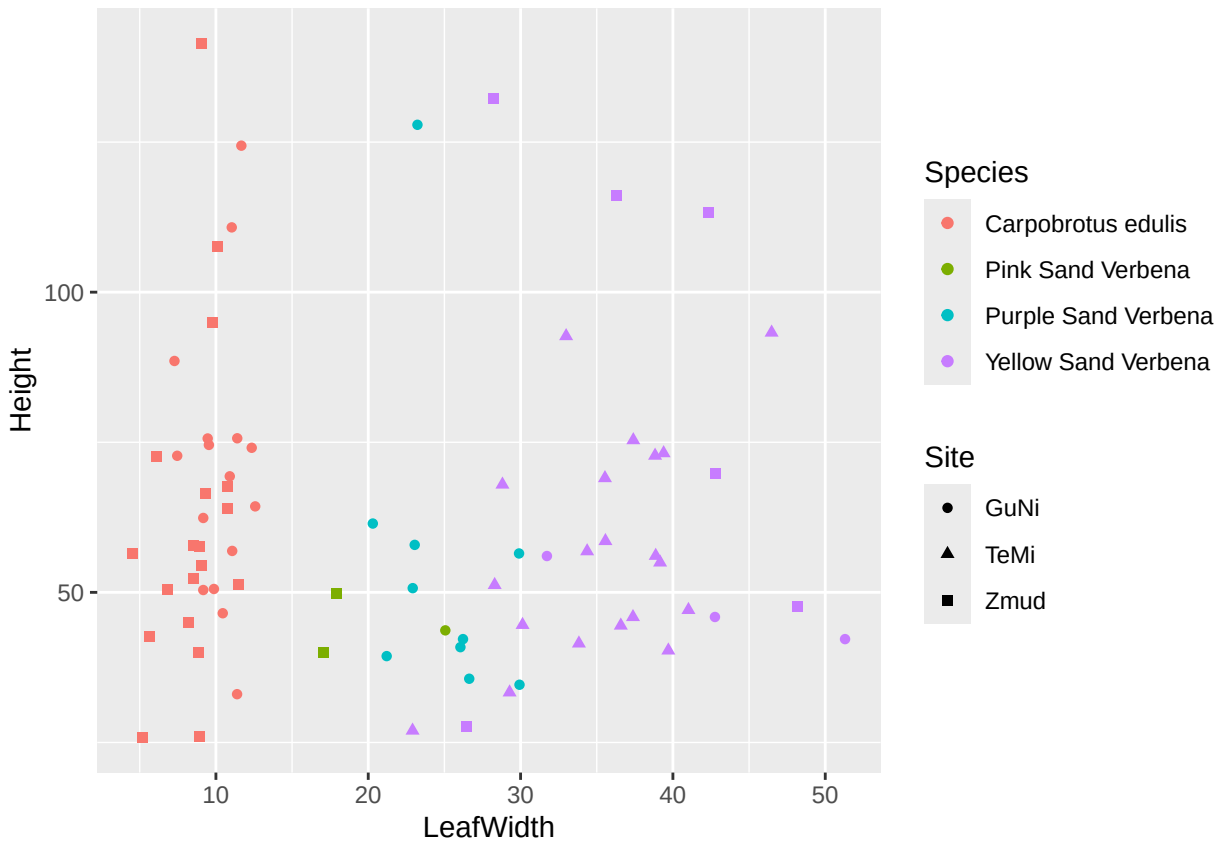

```
ggplot(aes(x = LeafLength, y = BasalDiam, color = Species, shape = Site), data = SuccTraits)+
  geom_point()
```



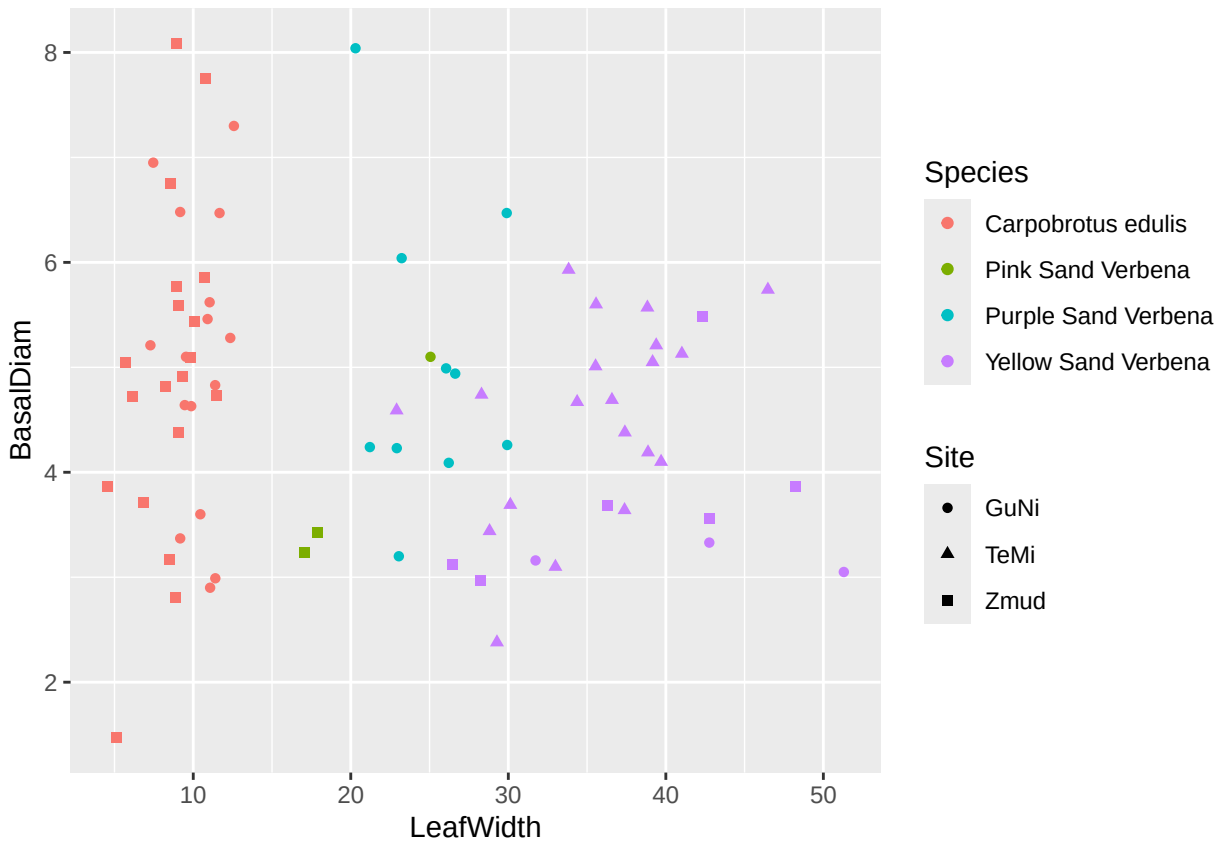

```
ggplot(aes(x = LeafWidth, y = LeafThickness, color = Species, shape = Site), data = SuccTraits)+
  geom_point()
```



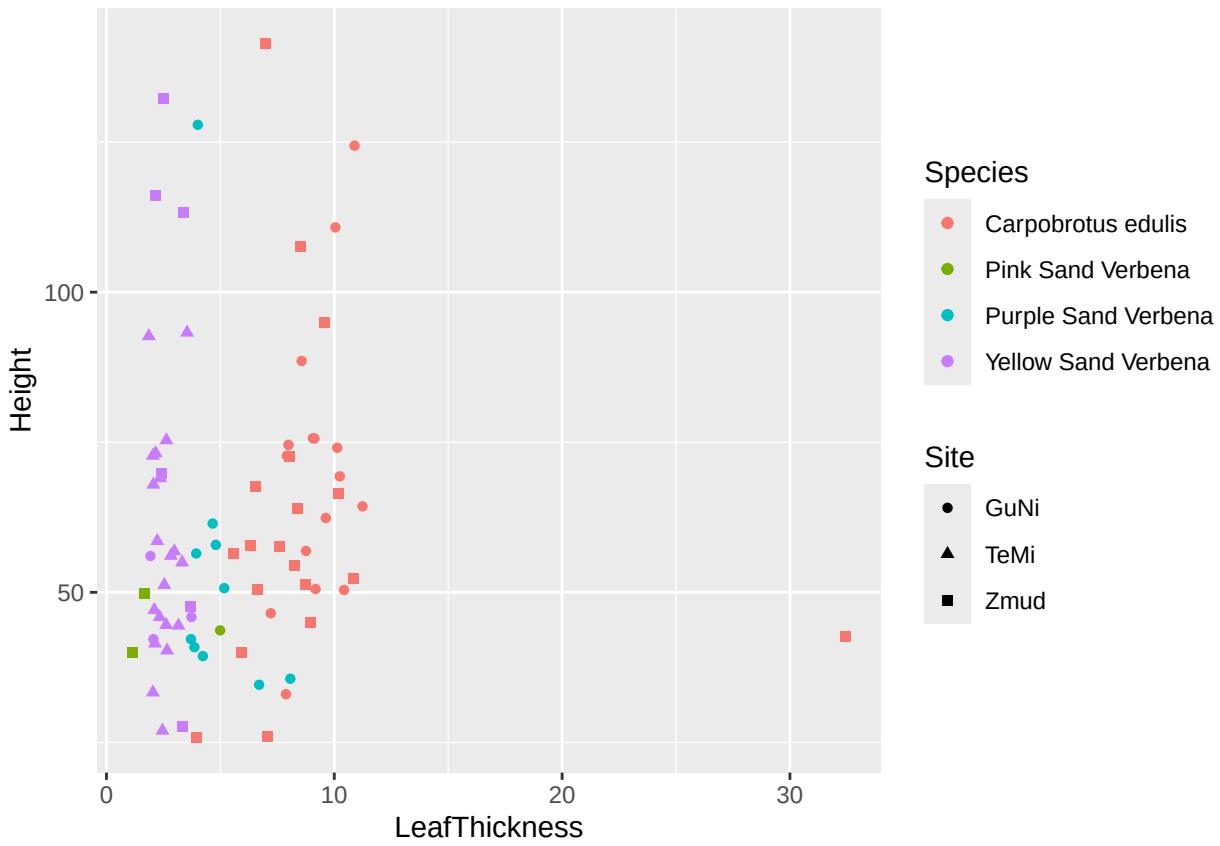

```
ggplot(aes(x = LeafWidth, y = Height, color = Species, shape = Site), data = SuccTraits)+
  geom_point()
```



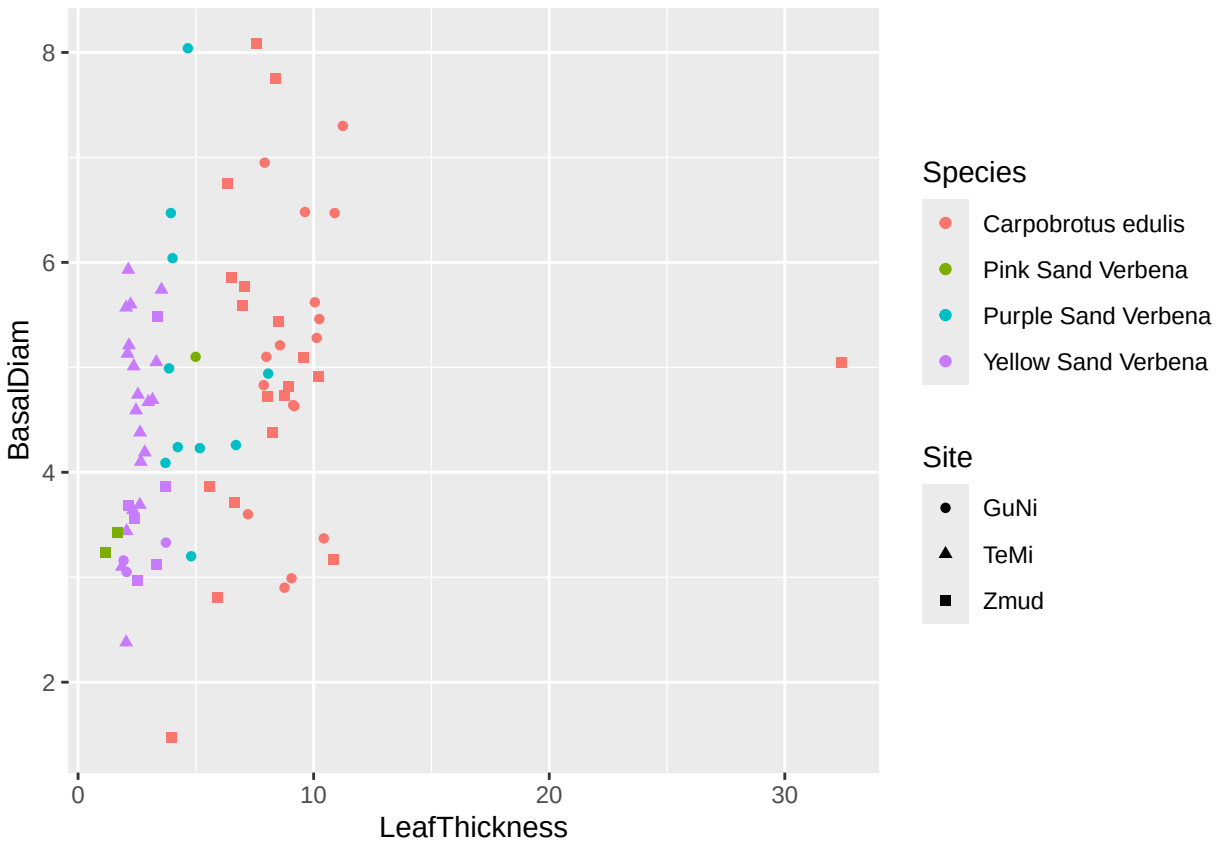

```
ggplot(aes(x = LeafWidth, y = BasalDiam, color = Species, shape = Site), data = SuccTraits)+
  geom_point()
```



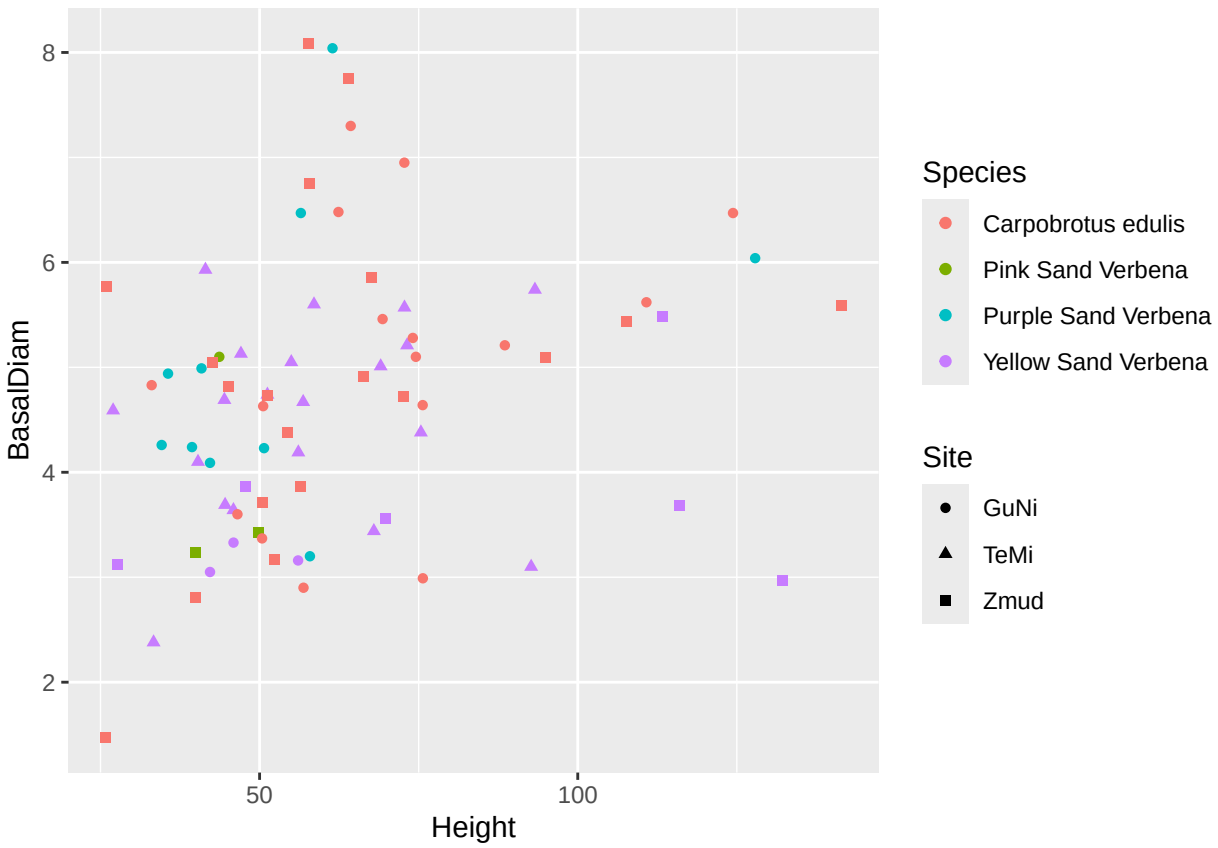

```
ggplot(aes(x = LeafThickness, y = Height, color = Species, shape = Site), data = SuccTraits)+
  geom_point()
```




```
ggplot(aes(x = LeafThickness, y = BasalDiam, color = Species, shape = Site), data = SuccTraits)+
  geom_point()
```




```
ggplot(aes(x = Height, y = BasalDiam, color = Species, shape = Site), data = SuccTraits)+
  geom_point()
```




```
save.image(file="DuneComm1.RData")
```